DISCRETE MATHEMATICS & GRAPH THEORY · Course Code 151304

# Unit 2

Principles of Mathematical Induction · Integers · Basic Counting Techniques

## Contents of this unit


**Part A Induction and Recursion**

- The Well-Ordering Principle
- Mathematical Induction (weak & strong)
- Recursive Definitions

**Part B Properties of Integers**

- The Division Algorithm
- Prime Numbers
- GCD and the Euclidean Algorithm
- Modular Arithmetic
- The Fundamental Theorem of Arithmetic

**Part C Basic Counting Techniques**

- The Rules of Sum and Product
- The Principle of Inclusion–Exclusion
- The Pigeon-hole Principle
- Permutations
- Combinations
- Binomial and Multinomial Theorems
- Combinations with Repetitions

**Part D Review**

- Unit 2 Question Bank
- Answers to Practice Questions


QP marks a frequently asked university question. Each topic ends with a **Practice Questions** box; answers are in Part D.


**Dr. Saurabh Suman**

Assistant Professor, Department of Computer Science & Engineering
Government Engineering College, Munger, Bihar


Lecture Notes

PART A
# Induction and Recursion

## 1 The Well-Ordering Principle and Mathematical Induction

The method of mathematical induction is based on a principle called the *Induction Principle*. This principle, in turn, is based on another principle known as the *Well-Ordering Principle*.

> **Note:** $\mathbb{Z}^+$ denotes the set of all positive integers; $\mathbb{N} = \{0, 1, 2, \dots\}$ denotes the set of natural numbers.

**Well-Ordering Principle** QP

"Every non-empty subset of $\mathbb{Z}^+$ contains a smallest (least) element."

(OR)

"The set of all positive integers is well ordered."

The principle looks obvious, but it is a powerful proof tool. Note that it fails for other sets: the set of all integers $\mathbb{Z}$, or the set of positive rationals $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$, has no least element.

**Induction Principle (Principle of Mathematical Induction)** QP

Let $S(n)$ denote an open statement that involves a positive integer $n$. Suppose that the following conditions hold:

(i) $S(1)$ is true.

(ii) Whenever $S(k)$ is true for some particular, arbitrarily chosen $k \in \mathbb{Z}^+$, then $S(k+1)$ is true.

Then $S(n)$ is true for all $n \in \mathbb{Z}^+$.

### The Well-Ordering Principle implies the Induction Principle NEW

**Proof.** Suppose (i) and (ii) hold, but $S(n)$ is *not* true for all $n \in \mathbb{Z}^+$. Then the set

$$F = \{\, n \in \mathbb{Z}^+ : S(n) \text{ is false} \,\}$$

is a non-empty subset of $\mathbb{Z}^+$. By the Well-Ordering Principle, $F$ has a least element, say $m$. Since $S(1)$ is true, $m \neq 1$, so $m \geq 2$ and $m - 1 \in \mathbb{Z}^+$. Because $m$ is the *least* element of $F$, $m - 1 \notin F$, i.e., $S(m-1)$ is true. By condition (ii) (with $k = m - 1$), $S(m)$ is true — contradicting $m \in F$. Hence $F = \varnothing$ and $S(n)$ is true $\forall n \in \mathbb{Z}^+$. ■

### Method of Mathematical Induction

Suppose we wish to prove that a certain statement $S(n)$ is true $\forall n \geq 1$. The method of proving this using the principle of induction is called the *method of mathematical induction*. There are two steps in this method:

1. **Basic step:** Verify that $S(n)$ is true for $n = 1$.
2. **Induction step:** Assuming $S(k)$ is true, where $k \geq 1$, show that $S(k+1)$ is true.

> **Note:**
> 1. Without the basic step, one cannot write the proof by induction.
> 2. Suppose we are required to prove a statement $S(n)$ for $n \geq n_0$, where $n_0$ is an integer. In such a situation, verifying that $S(n_0)$ is true is the basic step, and in the induction step we take $k \geq n_0$.

**Alternate Form of the Induction Principle (Strong Induction)**

Let $S(n)$ denote a statement that involves a positive integer $n$. Suppose that

(i) for some positive integers $n_0$, $n_0+1$, $n_0+2$, …, $n_0+p$, the statements $S(n_0)$, $S(n_0+1)$, …, $S(n_0+p)$

are true, and

(ii) if $S(n_0)$, $S(n_0+1)$, …, $S(k)$ are true, where $k \geq n_0 + p$, then $S(k+1)$ is true.

Then $S(n)$ is true $\forall n \geq n_0$.

**When to use which form?** Use ordinary (weak) induction when $S(k+1)$ can be obtained from $S(k)$ alone. Use the alternate (strong) form when $S(k+1)$ needs *several* earlier cases — e.g. recurrences like $a_n = a_{n-1} + a_{n-2}$, or factorisation into primes (Section 7).

## 1.1 Problems

**Problem 1.** Prove by mathematical induction that, $\forall n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \tfrac{1}{2}n(n+1).$$

**Solution.** Let $S(n):\ 1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1)$.

Basic step: $S(1)$ is the statement $1 = \frac{1}{2} \cdot 1 \cdot (2)$, which is true. Thus $S(n)$ is verified for $n = 1$.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., assume

$$S(k):\ 1 + 2 + 3 + \cdots + k = \tfrac{1}{2}k(k+1).$$

Adding $(k+1)$ to both sides, we get

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \tfrac{1}{2}k(k+1) + (k+1) \\ &= \tfrac{1}{2}(k+1)\,[\,k+2\,] = \tfrac{1}{2}(k+1)(k+2). \end{aligned}$$

This is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 2.** QP Prove by mathematical induction that

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \tfrac{1}{3}n(2n-1)(2n+1) \quad \forall n \geq 1.$$

**Solution.** Let $S(n):\ 1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$.

Basic step: $S(1):\ 1^2 = \frac{1}{3}(1)(1)(3)$, which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e.,

$$1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \tfrac{1}{3}k(2k-1)(2k+1).$$

Adding $(2k+1)^2$ on both sides (note: $[2(k+1)-1]^2 = (2k+1)^2$):

$$\begin{aligned} 1^2 + 3^2 + \cdots + (2k-1)^2 + (2k+1)^2 &= \tfrac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \\ &= \tfrac{1}{3}(2k+1)\,[\,k(2k-1) + 3(2k+1)\,] \\ &= \tfrac{1}{3}(2k+1)\,[\,2k^2 - k + 6k + 3\,] = \tfrac{1}{3}(2k+1)(2k^2+5k+3) \\ &= \tfrac{1}{3}(k+1)(2k+1)(2k+3). \end{aligned}$$

This is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 3.** Prove by mathematical induction that

$$1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \cdots + n(n+2) = \frac{n(n+1)(2n+7)}{6}.$$

**Solution.** Let $S(n)$ denote the given statement.

Basic step: $S(1):\ 1 \cdot 3 = \dfrac{1 \cdot 2 \cdot 9}{6} = 3$, which is true.

Induction step: Assume that $S(k)$ is true, where $k \geq 1$, i.e.,

$$1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \cdots + k(k+2) = \frac{k(k+1)(2k+7)}{6}.$$

Adding $(k+1)(k+3)$ on both sides, we get

$$\begin{aligned} 1 \cdot 3 + 2 \cdot 4 + \cdots + k(k+2) + (k+1)(k+3) &= \frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) \\ &= \frac{(k+1)}{6}\left[k(2k+7) + 6(k+3)\right] = \frac{(k+1)}{6}\left[2k^2 + 13k + 18\right] \\ &= \frac{(k+1)(k+2)(2k+9)}{6}. \end{aligned}$$

This is the statement $S(k+1)$, since $2(k+1)+7 = 2k+9$. So $S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 4.** QP Prove by mathematical induction that, for every positive integer $n$, 5 divides $n^5 - n$.

**Solution.**

> **Note:** "5 divides 30" means $\frac{30}{5}$ is an integer, i.e., 30 is a multiple of 5: $30 = 5 \times m$ for some integer $m$ (here $m = 6$).

Let $S(n)$ : 5 divides $n^5 - n$.

Basic step: $S(1)$ : 5 divides $1^5 - 1 = 0$ (i.e., $\frac{0}{5} = 0$), which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., 5 divides $k^5 - k$. So $k^5 - k$ is a multiple of 5: $k^5 - k = 5m$ for some integer $m$. Now

$$\begin{aligned} (k+1)^5 - (k+1) &= (k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1) - (k+1) \\ &= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k) \\ &= 5m + 5(k^4 + 2k^3 + 2k^2 + k) = 5(m + k^4 + 2k^3 + 2k^2 + k). \end{aligned}$$

$\Rightarrow (k+1)^5 - (k+1)$ is a multiple of 5, i.e., 5 divides $(k+1)^5 - (k+1)$. This is exactly the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, for $k \geq 1$. Therefore, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 5.** QP Prove that $4n < (n^2 - 7)$ for all positive integers $n \geq 6$.

**Solution.** Let $S(n)$ : $4n < (n^2 - 7)$. We prove that $S(n)$ is true for all positive integers $n \geq n_0$, where $n_0 = 6$.

Basic step: $S(6)$ : $(4 \times 6) < (6^2 - 7)$, i.e., $24 < 29$, which is true. So $S(n)$ is true for $n = n_0 = 6$.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 6$, i.e., $4k < (k^2 - 7)$ for $k \geq 6$. Adding 4 on both sides,

$$\begin{aligned} 4(k+1) = 4k + 4 &< (k^2 - 7) + 4 \\ &< (k^2 - 7) + (2k + 1), \end{aligned}$$

because when $k \geq 6$ we have $2k \geq 12 \Rightarrow 2k + 1 \geq 13 > 4$. Therefore

$$4(k+1) < (k^2 + 2k + 1) - 7, \quad \text{i.e.,} \quad 4(k+1) < (k+1)^2 - 7.$$

This is the statement $S(k+1)$. $\Rightarrow S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 6$. Thus $S(n)$ is true $\forall n \geq 6$. ■

**Problem 6.** Show that $2^n > n^2$ for all positive integers $n$ greater than 4.

**Solution.** We prove $S(n)$ : $2^n > n^2$ is true $\forall n \geq 5$.

Basic step: For $n = 5$: $S(5)$ : $2^5 > 5^2$ (i.e., $32 > 25$), which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 5$, i.e., $2^k > k^2$. Multiplying by 2 on both sides, $2 \cdot 2^k = 2^{k+1} > 2k^2$. Since $k > 4$, we have

$$\begin{aligned} 2k^2 = k^2 + k^2 = k^2 + (k \times k) &> k^2 + 4k = k^2 + 2k + 2k \\ &> k^2 + 2k + 1 = (k+1)^2. \end{aligned}$$

Thus $2^{k+1} > (k+1)^2$ for $k > 4$. This is precisely the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 5$. Hence, by mathematical induction, $S(n)$ is true $\forall n > 4$. ■

**Problem 7.** By mathematical induction, prove that $n! \geq 2^{n-1}\ \forall n \geq 1$.

**Solution.** Let $S(n)$ be the given statement.

Basic step: For $n = 1$: $S(1):\ 1! \geq 2^{1-1}$, i.e., $1 \geq 1$, which is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 1$, i.e., $k! \geq 2^{k-1}$, i.e., $2^{k-1} \leq k!$. Multiplying by 2 on both sides,

$$\begin{aligned} 2^k = 2 \cdot 2^{k-1} &\leq 2 \cdot k! \\ &\leq (k+1) \cdot k! \qquad \text{because } (k+1) \geq 2 \text{ for } k \geq 1 \\ &= (k+1)! \end{aligned}$$

i.e., $(k+1)! \geq 2^k$, which is the statement $S(k+1)$. Thus $S(k+1)$ is true whenever $S(k)$ is true, for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 8.** For all $n \in \mathbb{Z}^+$, show that if $n \geq 14$, then $n$ can be written as a sum of 3's and/or 8's.

**Solution.** To prove: $S(n):\ n$ can be written as a sum of 3's and/or 8's, is true for all integers $n \geq 14$.

Basic step: We have $14 = (3+3)+8$. $\Rightarrow S(14)$ is true.

Induction step: Assume that $S(n)$ is true for $n = k$, where $k \geq 14$. Then $k = (3+3+3+\cdots)+(8+8+\cdots)$. Suppose this representation of $k$ has $r$ number of 3's and $s$ number of 8's.

*Case 1: $s \geq 1$ (at least one 8 is used).* Then

$$\begin{aligned} k+1 &= \{\underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s}\} + 1 = \underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s-1} + 8 + 1 \\ &= \underbrace{(3+3+\cdots)}_{r} + \underbrace{(8+8+\cdots)}_{s-1} + (3+3+3) = \underbrace{(3+3+\cdots)}_{r+3} + \underbrace{(8+8+\cdots)}_{s-1}. \end{aligned}$$

*Case 2: $s = 0$ (only 3's are used).* Then $k = 3r \geq 14$, so $r \geq 5$. Replacing five 3's (sum 15) by two 8's (sum 16),

$$k+1 = \underbrace{(3+3+\cdots)}_{r-5} + (8+8).$$

$\Rightarrow (k+1)$ is a sum of 3's and/or 8's, i.e., $S(k+1)$ is true. Hence, by mathematical induction, $S(n)$ is true for all positive integers $n \geq 14$. ■

**Problem 9.** By mathematical induction, prove that for every positive integer $n$, the number $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8.

**Solution.** Let $S(n)$ be the given statement.

Basic step: $S(1):\ 5^1 + 2 \cdot 3^0 + 1 = 8$ is a multiple of 8. So, for $n = 1$, $S(n)$ is true.

Induction step: Assume that $A_n$ is a multiple of 8 for $n = k \geq 1$, i.e., $A_k = 5^k + 2 \cdot 3^{k-1} + 1$ is a multiple of 8. We find

$$\begin{aligned} A_{k+1} - A_k &= (5^{k+1} + 2 \cdot 3^k + 1) - (5^k + 2 \cdot 3^{k-1} + 1) \\ &= (5^k \cdot 5 - 5^k) + 2(3^k - 3^{k-1}) = 5^k(5-1) + 2 \cdot 3^{k-1}(3-1) \\ &= 4 \cdot 5^k + 4 \cdot 3^{k-1} = 4(5^k + 3^{k-1}). \end{aligned}$$

5 and 3 are odd $\Rightarrow 5^k$ and $3^{k-1}$ are also odd $\Rightarrow 5^k + 3^{k-1}$ is even $\Rightarrow 4(5^k + 3^{k-1})$ is a multiple of 8, i.e., $A_{k+1} - A_k$ is a multiple of 8. Since $A_k$ is a multiple of 8 (by assumption), $A_{k+1}$ is also a multiple of 8. Hence, by mathematical induction, for every positive integer $n$, $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8. ■

**Problem 10.** QP Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer $n$.

**Solution.** Let $A_n = 6^{n+2} + 7^{2n+1}$.

Basic step: $A_1 = 6^3 + 7^3 = 216 + 343 = 559 = 43 \times 13$ is divisible by 43. So $A_n$ is divisible by 43 for $n = 1$.

Induction step: Assume that $A_n$ is divisible by 43 for $n = k \geq 1$, i.e., $A_k = 6^{k+2} + 7^{2k+1}$ is divisible by 43. Then

$$\begin{aligned} A_{k+1} &= 6^{k+3} + 7^{2(k+1)+1} = 6^{k+3} + 7^{2k+3} = 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 7^2 \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 49 = 6^{k+2} \cdot 6 + 7^{2k+1}(6 + 43) \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 6 + 7^{2k+1} \cdot 43 \qquad [\,a(b+c) = ab + ac\,] \\ &= 6\,[\,6^{k+2} + 7^{2k+1}\,] + 43 \cdot 7^{2k+1} = 6A_k + 43 \cdot 7^{2k+1}. \end{aligned}$$

By assumption, $A_k$ is divisible by 43 $\Rightarrow 6A_k$ is also divisible by 43; and $43 \cdot 7^{2k+1}$ is divisible by 43. Therefore $A_{k+1}$ is divisible by 43. Thus if $S(k)$ is true, then $S(k+1)$ is true, for $k \geq 1$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 11.** QP Prove that $\displaystyle\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$ $\forall n \in \mathbb{Z}^+$, i.e.,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

**Solution.** Let $S(n)$ be the given statement.

Basic step: $S(1)$ : $\dfrac{1}{1(2)} = \dfrac{1}{1+1} \Rightarrow \dfrac{1}{2} = \dfrac{1}{2}$, which is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e., $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \cdots + \dfrac{1}{k(k+1)} = \dfrac{k}{k+1}$. Adding $\dfrac{1}{(k+1)(k+2)}$ on both sides,

$$\begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(k+1)(k+2)} &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{1}{k+1}\left[k + \frac{1}{k+2}\right] \\ &= \frac{1}{k+1}\left[\frac{k^2 + 2k + 1}{k+2}\right] = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}, \end{aligned}$$

which is precisely the statement $S(k+1)$. Hence, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 12.** QP Establish the following by mathematical induction: $\displaystyle\sum_{i=1}^{n} i\,2^i = 2 + (n-1)2^{n+1}$.

**Solution.** Let $S(n)$ be the given statement.

Basic step: $S(1)$ : $1 \times 2^1 = 2 + (1-1)2^2$, i.e., $2 = 2$, which is true.

Induction step: Assume that the result is true for $n = k$, where $k \geq 1$, i.e.,

$$S(k) : \; 1 \times 2^1 + 2 \times 2^2 + 3 \times 2^3 + \cdots + k \times 2^k = 2 + (k-1)2^{k+1}.$$

Adding $(k+1) \times 2^{k+1}$ on both sides,

$$\begin{aligned} 1 \times 2^1 + 2 \times 2^2 + \cdots + k \times 2^k + (k+1) \times 2^{k+1} &= 2 + (k-1)2^{k+1} + (k+1)2^{k+1} \\ &= 2 + 2^{k+1}[\,k - 1 + k + 1\,] = 2 + 2^{k+1} \cdot 2k = 2 + k\,2^{k+2}, \end{aligned}$$

which is precisely the statement $S(k+1)$. Hence $S(n)$ is true $\forall n \geq 1$. ■

**Homework**

Prove by mathematical induction:

1. $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1)$.
2. $1^3 + 2^3 + 3^3 + \cdots + n^3 = \dfrac{n^2(n+1)^2}{4}$.

3. $2^n < n!$ for all positive integers $n > 3$. QP
4. For every integer $n$, 3 divides $n^3 - n$.
5. For all $n \in \mathbb{Z}^+$, show that if $n \geq 24$, then $n$ can be written as a sum of 5's and/or 7's. QP

**Problem 13.** QP Let $H_1 = 1,\ H_2 = 1 + \frac{1}{2},\ H_3 = 1 + \frac{1}{2} + \frac{1}{3},\ \ldots,\ H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$. Prove that

$$\sum_{i=1}^{n} H_i = (n+1)H_n - n, \quad \forall n \geq 1.$$

**Solution.** Let $S(n) : \sum_{i=1}^{n} H_i = (n+1)H_n - n$, i.e., $H_1 + H_2 + H_3 + \cdots + H_n = (n+1)H_n - n$.

Basic step: $S(1) : H_1 = (1+1)H_1 - 1$, i.e., $1 = 2(1) - 1$, i.e., $1 = 1$. $\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e., $H_1 + H_2 + \cdots + H_k = (k+1)H_k - k$. Adding $H_{k+1}$ on both sides,

$$\begin{aligned} H_1 + H_2 + \cdots + H_k + H_{k+1} &= (k+1)H_k - k + H_{k+1} \\ &= (k+1)\left[H_{k+1} - \frac{1}{k+1}\right] - k + H_{k+1} \\ &\quad \left[\because H_{k+1} = H_k + \tfrac{1}{k+1},\ \text{so } H_k = H_{k+1} - \tfrac{1}{k+1}\right] \\ &= (k+1)H_{k+1} - 1 - k + H_{k+1} = (k+2)H_{k+1} - (k+1). \end{aligned}$$

This is precisely the statement $S(k+1)$. By mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

## 1.2 Additional Examples NEW

**Example 1.** (*A proof using the Well-Ordering Principle directly.*) Prove that every integer $n \geq 2$ has at least one prime divisor.

**Solution.** Suppose not. Then the set $C = \{n \geq 2 : n \text{ has no prime divisor}\}$ is non-empty, so by the Well-Ordering Principle it has a least element $m$. Now $m$ itself is not prime (a prime is its own prime divisor). So $m = ab$ with $1 < a < m$. Since $a < m$ and $a \geq 2$, $a \notin C$, so $a$ has a prime divisor $p$. But $p \mid a$ and $a \mid m$ give $p \mid m$ — a contradiction. Hence $C = \varnothing$. ■

**Example 2.** (*Strong induction.*) Prove that every amount of postage of 12 paise or more can be formed using only 4-paise and 5-paise stamps.

**Solution.** Let $S(n)$: postage of $n$ paise can be formed with 4p and 5p stamps.

Basic step: We verify four consecutive cases ($p = 3$ in the alternate form): $12 = 4+4+4$, $13 = 4+4+5$, $14 = 4+5+5$, $15 = 5+5+5$.

Induction step: Assume $S(12), S(13), \ldots, S(k)$ are true for some $k \geq 15$. Then $k+1-4 = k-3 \geq 12$, so $S(k-3)$ is true. Adding one 4-paise stamp to the stamps for $k-3$ paise gives postage of $k+1$ paise. So $S(k+1)$ is true. By the alternate form of induction, $S(n)$ is true $\forall n \geq 12$. ■

**Example 3.** Prove by induction that $3^n > n^3$ for all integers $n \geq 4$. Hint for the induction step is built in below.

**Solution.** Basic step: $3^4 = 81 > 64 = 4^3$. True.

Induction step: Assume $3^k > k^3$ for some $k \geq 4$. Then $3^{k+1} = 3 \cdot 3^k > 3k^3$. We need $3k^3 \geq (k+1)^3$, i.e., $\left(1 + \frac{1}{k}\right)^3 \leq 3$. For $k \geq 4$: $\left(1 + \frac{1}{4}\right)^3 = \frac{125}{64} < 2 < 3$, and $\left(1 + \frac{1}{k}\right)^3$ decreases as $k$ increases. Hence $3^{k+1} > (k+1)^3$. By induction, the result holds $\forall n \geq 4$. ■

**Example 4.** (*A common error.*) What is wrong with the following "proof" that all horses have the same colour? *"$S(1)$ is clearly true. Assume any $k$ horses have the same colour. Given $k+1$ horses, remove horse 1: the remaining $k$ have the same colour. Remove horse $k+1$: the remaining $k$ have the same colour. The two groups overlap, so all $k+1$ have the same colour."*

**Solution.** The induction step fails for $k = 1$: with 2 horses, the two groups $\{h_2\}$ and $\{h_1\}$ *do not overlap*. So "$S(1) \Rightarrow S(2)$" is never proved. Lesson: the induction step must be valid for *every* $k \geq n_0$, including the smallest one.

**Practice Questions — Induction**

1. Prove that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all $n \geq 1$.
2. Prove that $2 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 2$ for all $n \geq 1$.
3. Prove that $\dfrac{1}{1\cdot 3} + \dfrac{1}{3\cdot 5} + \cdots + \dfrac{1}{(2n-1)(2n+1)} = \dfrac{n}{2n+1}$.
4. Prove that $11^{n+2} + 12^{2n+1}$ is divisible by 133 for every $n \geq 0$. QP
5. Prove that $7^n - 1$ is divisible by 6 for all $n \geq 1$.
6. Prove that $n^3 + 2n$ is divisible by 3 for all $n \geq 1$.
7. Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$.
8. Prove that every postage of 8 paise or more can be made using 3-paise and 5-paise stamps (use strong induction).
9. Using the Well-Ordering Principle, prove that there is no integer strictly between 0 and 1.

# 2 Recursive Definitions

Consider $a_1, a_2, \ldots, a_n, \ldots$, a sequence in which $a_1$ is the first term, $a_2$ is the 2nd term, $\ldots$, $a_n$ is the $n^{\text{th}}$ term. The element $a_n$ is called the *general term* of the sequence, and the sequence is denoted by $\{a_n\}$. If the first term is denoted by $a_0$, then the $n^{\text{th}}$ term is denoted by $a_{n-1}$ and the sequence is denoted by $\{a_{n-1}\}$.

For describing a sequence, two methods are commonly used: (i) the explicit method, (ii) the recursive method.

In the **explicit method**, the general term of the sequence is explicitly indicated.
*Example:* $E = \{a_n\}$, where $a_n = 2n$, $n \in \mathbb{Z}^+$.

In the **recursive method**, the first few terms of the sequence are explicitly indicated and the general term is specified through a rule (formula) which indicates how to obtain new terms of the sequence from the terms already known.
*Example:* Fibonacci numbers: $F_0 = 0,\ F_1 = 1,\ F_n = F_{n-1} + F_{n-2}\ \forall n \geq 2$.

**Note:**

1. A recursive definition of a sequence should contain two parts: in the 1st part (the *basis*), a few terms of the sequence are indicated explicitly, and in the 2nd part (the *recursive rule*), the rule to obtain new terms of the sequence is indicated.
2. Recursive rule/formula: a formula for $a_n$ in terms of $a_{n-1}, a_{n-2}, \ldots$ (or) a formula for $a_{n+1}$ in terms of $a_n, a_{n-1}, \ldots$

## 2.1 Recursively Defined Functions and Sets NEW

The same two-part idea is used to define *functions* and *sets*. Recursion is the mathematical basis of recursive programs, and induction is the tool for proving them correct.

**Recursively defined functions**

| | |
|---|---|
| Factorial: | $0! = 1;\quad n! = n \cdot (n-1)!\ \ (n \geq 1)$ |
| Power: | $a^0 = 1;\quad a^n = a \cdot a^{n-1}\ \ (n \geq 1)$ |
| Sum: | $\sum_{i=1}^{1} a_i = a_1;\quad \sum_{i=1}^{n} a_i = \left(\sum_{i=1}^{n-1} a_i\right) + a_n$ |
| GCD: | $\gcd(a, 0) = a;\quad \gcd(a, b) = \gcd(b,\ a \bmod b)\ \ (b > 0)$ (see Section 5) |

**Recursively defined sets**

A set $S$ is defined recursively by (i) a *basis step* listing some initial elements, and (ii) a *recursive step* giving rules to form new elements from known ones. Only elements obtained this way belong to $S$.
*Example:* $3 \in S$; if $x \in S$ and $y \in S$ then $x + y \in S$. Then $S = \{3, 6, 9, 12, \ldots\}$, the positive multiples of 3.

*Example (strings):* Let $\Sigma$ be an alphabet. $\lambda$ (the empty string) $\in \Sigma^*$; if $w \in \Sigma^*$ and $x \in \Sigma$, then $wx \in \Sigma^*$.

**Example 1.** Give a recursive definition of $f(n) = n!$ and use it to compute 5!.

**Solution.** $f(0) = 1$, $f(n) = n \cdot f(n-1)$. Then $f(1) = 1$, $f(2) = 2$, $f(3) = 6$, $f(4) = 24$, $f(5) = 5 \cdot 24 = 120$.

**Example 2.** (*Tower of Hanoi.*) Let $H_n$ be the minimum number of moves needed to transfer $n$ discs. Then $H_1 = 1$ and $H_n = 2H_{n-1} + 1$ (move $n-1$ discs aside, move the largest disc, move the $n-1$ discs back on top). Prove that $H_n = 2^n - 1$.

**Solution.** Basic step: $H_1 = 1 = 2^1 - 1$. Induction step: assume $H_k = 2^k - 1$. Then $H_{k+1} = 2H_k + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 1$. Hence $H_n = 2^n - 1 \; \forall n \geq 1$. (For 64 discs this is about $1.8 \times 10^{19}$ moves.) ■

## 2.2 Problems

**Problem 1.** Obtain a recursive definition for the sequence $\{a_n\}$ in each of the following cases: (i) $a_n = 6^n$ (ii) $a_n = n(n+3)$ (iii) $a_n = (n+1)!$ (iv) $a_n = 2 - (-1)^n$ QP

**Solution.** (i) $a_n = 6^n$.

$$a_1 = 6^1 = 6, \quad a_2 = 6^2 = 36 = 6 \times a_1, \quad a_3 = 6^3 = 216 = 6 \times a_2, \quad a_4 = 6^4 = 1296 = 6 \times a_3, \ \ldots$$

$a_1 = 6$; $a_n = 6a_{n-1} \; \forall n \geq 2$ is the recursive definition (same as $a_{n+1} = 6a_n \; \forall n \geq 1$).

(ii) $a_n = n(n+3)$.

$$a_1 = 1(1+3) = 4, \qquad a_2 = 2 \times 5 = 10 = a_1 + 6 = a_1 + (2 \times 3),$$
$$a_3 = 3 \times 6 = 18 = a_2 + (2 \times 4), \qquad a_4 = 4 \times 7 = 28 = a_3 + (2 \times 5), \ \ldots$$

Thus $a_1 = 4$, $a_n = a_{n-1} + 2(n+1) \; \forall n \geq 2$ is the recursive definition.

(iii) $a_n = (n+1)!$.

$$a_1 = 2! = 2, \quad a_2 = 3! = 6 = a_1 \times 3, \quad a_3 = 4! = 24 = a_2 \times 4, \quad a_4 = 5! = 120 = a_3 \times 5$$

$\Rightarrow a_n = a_{n-1} \times (n+1) \; \forall n \geq 2$. Thus $a_1 = 2$, $a_n = (n+1)a_{n-1} \; \forall n \geq 2$ is the recursive definition.

(iv) $a_n = 2 - (-1)^n$.

$$a_1 = 2 - (-1)^1 = 3, \qquad a_2 = 2 - (-1)^2 = 1 = 3 - 2 = a_1 - (-1)^2 \times 2,$$
$$a_3 = 2 - (-1)^3 = 3 = 1 + 2 = a_2 - (-1)^3 \times 2, \qquad a_4 = 1 = 3 - 2 = a_3 - (-1)^4 \times 2, \ \ldots$$

$a_1 = 3$, $a_n = a_{n-1} - 2(-1)^n \; \forall n \geq 2$ is the recursive definition.

**Problem 2.** QP Find an explicit definition of the sequence defined recursively by $a_1 = 7$, $a_n = 2a_{n-1} + 1$ for $n \geq 2$.

**Solution.**

$$\begin{aligned} a_n &= 2a_{n-1} + 1 = 2^1 a_{n-1} + (2^1 - 1) \\ &= 2\{2a_{n-2} + 1\} + 1 = 4a_{n-2} + 3 = 2^2 a_{n-2} + (2^2 - 1) \\ &= 4\{2a_{n-3} + 1\} + 3 = 8a_{n-3} + 7 = 2^3 a_{n-3} + (2^3 - 1) \\ &= 8\{2a_{n-4} + 1\} + 7 = 16a_{n-4} + 15 = 2^4 a_{n-4} + (2^4 - 1) \\ &\vdots \end{aligned}$$

(Observe: the power of 2 and the subscript add up to $n$; when the subscript becomes 1, the power is $n-1$.)

$$\begin{aligned} \therefore a_n &= 2^{n-1} \cdot a_1 + 2^{n-1} - 1 = 2^{n-1}\{a_1 + 1\} - 1 = 2^{n-1}\{7 + 1\} - 1 \\ &= 2^{n-1} \cdot 2^3 - 1 = 2^{n+2} - 1 = 4 \times 2^n - 1. \end{aligned}$$

Thus $a_n = 4 \times 2^n - 1$ is an explicit definition for the given sequence.

**Problem 3.** QP A sequence $\{a_n\}$ is defined recursively by $a_1 = 4$, $a_n = a_{n-1} + n\ \forall n \geq 2$. Find $a_n$ in explicit form.

**Solution.**

$$\begin{aligned} a_n &= a_{n-1} + n = \{a_{n-2} + (n-1)\} + n = a_{n-3} + (n-2) + (n-1) + n \\ &\vdots \\ &= a_1 + 2 + 3 + 4 + \cdots + n = (a_1 - 1) + (1 + 2 + 3 + \cdots + n) = (a_1 - 1) + \frac{n(n+1)}{2}. \end{aligned}$$

$a_n = 3 + \dfrac{n(n+1)}{2}$ is the explicit formula for $a_n$.

**Problem 4.** QP A sequence $\{a_n\}$ is defined recursively by $a_0 = 1,\ a_1 = 1,\ a_2 = 1$ and $a_n = a_{n-1} + a_{n-3}\ \forall n \geq 3$. Prove that $a_{n+2} \geq (\sqrt{2})^n\ \forall n \geq 0$.

**Solution.** Let $S(n):\ a_{n+2} \geq (\sqrt{2})^n$.

Basic step: First, $a_3 = a_2 + a_0 = 1 + 1 = 2$ and $a_4 = a_3 + a_1 = 2 + 1 = 3$.

$$S(0):\ a_2 \geq (\sqrt{2})^0,\ \text{i.e., } 1 \geq 1; \quad S(1):\ a_3 \geq (\sqrt{2})^1,\ \text{i.e., } 2 \geq \sqrt{2}; \quad S(2):\ a_4 \geq (\sqrt{2})^2,\ \text{i.e., } 3 \geq 2.$$

Thus $S(n)$ is true for $n = 0, 1, 2$.

Induction step: Assume (alternate form of induction) that $S(0), S(1), \ldots, S(k)$ are true for some $k \geq 2$; in particular

$$a_{k+2} \geq (\sqrt{2})^k \quad \text{and} \quad a_k \geq (\sqrt{2})^{k-2}. \tag{1}$$

To prove: $a_{k+3} \geq (\sqrt{2})^{k+1}$. Consider

$$\begin{aligned} a_{k+3} &= a_{k+2} + a_k \qquad \text{(using the definition of } a_n) \\ &\geq (\sqrt{2})^k + (\sqrt{2})^{k-2} \qquad \text{(using (1))} \\ &= (\sqrt{2})^{k-2}\left[(\sqrt{2})^2 + 1\right] = (\sqrt{2})^{k-2}\,[3] = (\sqrt{2})^{k+1} \cdot (\sqrt{2})^{-3} \cdot 3 = (\sqrt{2})^{k+1} \cdot \frac{3}{2\sqrt{2}} \\ &\geq (\sqrt{2})^{k+1} \qquad \left[\because \tfrac{3}{2\sqrt{2}} \geq 1\right]. \end{aligned}$$

Hence, by mathematical induction, the required result is true $\forall n \geq 0$. ■

**Problem 5.** QP For the sequence $\{a_n\}$ defined recursively by $a_1 = 8,\ a_2 = 22,\ a_n = 4(a_{n-1} - a_{n-2})$ for $n \geq 3$, prove that $a_n = (5 + 3n)2^{n-1}$ for $n \geq 1$.

**Solution.** $S(n):\ a_n = (5 + 3n)2^{n-1}\ \forall n \geq 1$.

Basic step: $S(1):\ a_1 = (5 + 3)2^0 = 8$. $S(2):\ a_2 = (5 + 6)2 = 22$. So $S(n)$ is true for $n = 1, 2$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$, i.e.,

$$a_k = (5 + 3k)2^{k-1}, \qquad a_{k-1} = \{5 + 3(k-1)\}2^{k-2}. \tag{1}$$

To prove: $a_{k+1} = (8 + 3k)2^k$. Consider

$$\begin{aligned} a_{k+1} &= 4(a_k - a_{k-1}) \qquad \text{(using the definition of } a_n) \\ &= 4\left[(5 + 3k)2^{k-1} - \{5 + 3(k-1)\}2^{k-2}\right] \qquad \text{(using (1))} \\ &= 4\left[5(2^{k-1} - 2^{k-2}) + 3\{k \cdot 2^{k-1} - (k-1)2^{k-2}\}\right] \\ &= 4\left[5 \cdot 2^{k-2}(2-1) + 3 \cdot 2^{k-2}\{2k - k + 1\}\right] = 4\left[5 \cdot 2^{k-2} + 3 \cdot 2^{k-2}(k+1)\right] \\ &= 4 \cdot 2^{k-2}[5 + 3k + 3] = 2^2 \cdot 2^{k-2}[8 + 3k] = 2^k(8 + 3k), \end{aligned}$$

which is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

**Problem 6.** QP A sequence $\{c_n\}$ is defined recursively by $c_n = 3c_{n-1} - 2c_{n-2}\ \forall n \geq 3$, with $c_1 = 5$ and $c_2 = 3$. Show that $c_n = -2^n + 7$.

**Solution.** Let $S(n):\ c_n = -2^n + 7\ \forall n \geq 1$.

Basic step: $S(1):\ c_1 = -2^1 + 7$, i.e., $5 = 5$. $S(2):\ c_2 = -2^2 + 7$, i.e., $3 = 3$. $\therefore S(n)$ is true for $n = 1, 2$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$: $c_k = -2^k + 7$ and $c_{k-1} = -2^{k-1} + 7$. To prove: $c_{k+1} = -2^{k+1} + 7$. Consider

$$\begin{aligned} c_{k+1} &= 3c_k - 2c_{k-1} = 3(-2^k + 7) - 2(-2^{k-1} + 7) \\ &= (-3 \cdot 2^k + 2 \cdot 2^{k-1}) + 21 - 14 = 2^{k-1}(-3 \cdot 2 + 2) + 7 \\ &= -4 \cdot 2^{k-1} + 7 = -2^2 \cdot 2^{k-1} + 7 = -2^{k+1} + 7, \end{aligned}$$

which is exactly the statement $S(k+1)$. Thus, by mathematical induction, $S(n)$ is true $\forall n \geq 1$. ■

**Problem 7.** QP The Fibonacci numbers are defined by $F_0 = 0,\ F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}\ \forall n \geq 2$. Prove that $\sum_{i=0}^{n} F_i^2 = F_n \times F_{n+1}$.

**Solution.** Let $S(n):\ \sum_{i=0}^{n} F_i^2 = F_n \times F_{n+1}$.

Basic step: $S(0):\ F_0^2 = F_0 \times F_1$, i.e., $0^2 = 0 \times 1$, i.e., $0 = 0$.
$S(1):\ F_0^2 + F_1^2 = F_1 \times F_2$, i.e., $0 + 1^2 = 1 \times (F_1 + F_0) = 1 \times (1 + 0)$, i.e., $1 = 1$. $S(n)$ is true for $n = 0, 1$.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e., $\sum_{i=0}^{k} F_i^2 = F_k \times F_{k+1}$ (1)
To prove: $\sum_{i=0}^{k+1} F_i^2 = F_{k+1} \times F_{k+2}$. Consider

$$\begin{aligned} \sum_{i=0}^{k+1} F_i^2 &= \sum_{i=0}^{k} F_i^2 + F_{k+1}^2 = (F_k \times F_{k+1}) + F_{k+1}^2 \qquad \text{(using (1))} \\ &= F_{k+1}(F_k + F_{k+1}) = F_{k+1}(F_{k+2}) \qquad (\because F_n = F_{n-1} + F_{n-2}). \end{aligned}$$

This is exactly the statement $S(k+1)$. Thus, by mathematical induction, $S(n)$ is true $\forall n \geq 0$. ■

**Problem 8.** QP For the Fibonacci sequence $F_0, F_1, F_2, \ldots$ prove that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

**Solution.** Let $S(n)$ denote the given statement, and write $\alpha = \dfrac{1+\sqrt{5}}{2},\ \beta = \dfrac{1-\sqrt{5}}{2}$.

Basic step: $S(0):\ F_0 = \frac{1}{\sqrt{5}}[1 - 1] = 0$. $\quad S(1):\ F_1 = \dfrac{1}{\sqrt{5}}\left[\dfrac{1}{2} + \dfrac{\sqrt{5}}{2} - \dfrac{1}{2} + \dfrac{\sqrt{5}}{2}\right] = \dfrac{1}{\sqrt{5}} \cdot \dfrac{2\sqrt{5}}{2} = 1$. Thus $S(n)$ is true for $n = 0, 1$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 1$, i.e., $F_k = \frac{1}{\sqrt{5}}[\alpha^k - \beta^k]$ and $F_{k-1} = \frac{1}{\sqrt{5}}[\alpha^{k-1} - \beta^{k-1}]$.
To prove: $F_{k+1} = \frac{1}{\sqrt{5}}[\alpha^{k+1} - \beta^{k+1}]$. Consider $F_{k+1} = F_k + F_{k-1}$ (by definition).

$$\begin{aligned} \Rightarrow F_{k+1} &= \frac{1}{\sqrt{5}}\left[\alpha^k - \beta^k\right] + \frac{1}{\sqrt{5}}\left[\alpha^{k-1} - \beta^{k-1}\right] = \frac{1}{\sqrt{5}}\left[\alpha^{k-1}\{\alpha + 1\} - \beta^{k-1}\{\beta + 1\}\right] \\ &= \frac{1}{\sqrt{5}}\left[\alpha^{k-1}\left(\frac{3+\sqrt{5}}{2}\right) - \beta^{k-1}\left(\frac{3-\sqrt{5}}{2}\right)\right] = \frac{1}{\sqrt{5}}\left[\alpha^{k-1}\alpha^2 - \beta^{k-1}\beta^2\right] \end{aligned}$$

$$\left[\because \alpha^2 = \frac{1+5+2\sqrt{5}}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2};\ \text{similarly } \beta^2 = \frac{3-\sqrt{5}}{2}\right]$$

$$\therefore F_{k+1} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right].$$

Thus $S(k+1)$ is true whenever $S(k-1)$ and $S(k)$ are true. By mathematical induction, the result is true $\forall n \geq 0$. ■

**Problem 9.** QP Let $F_n$ denote the Fibonacci numbers. Prove that $\sum_{i=1}^{n} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{n+2}}{2^n}$.

**Solution.** Let $S(n)$ denote the given statement.

Basic step: For $n = 1$: $S(1): \ \frac{F_0}{2^1} = 1 - \frac{F_3}{2^1}$. We have $F_0 = 0,\ F_1 = 1,\ F_2 = 1,\ F_3 = 2$. $\therefore S(1): \ \frac{0}{2} = 1 - \frac{2}{2} \Rightarrow 0 = 0$. True.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e., $\sum_{i=1}^{k} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{k+2}}{2^k}$. (1)

To prove: $\sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = 1 - \frac{F_{k+3}}{2^{k+1}}$. Consider

$$\sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = \underbrace{\frac{F_0}{2} + \frac{F_1}{2^2} + \cdots + \frac{F_{k-1}}{2^k}}_{\sum_{i=1}^{k} F_{i-1}/2^i} + \frac{F_k}{2^{k+1}} = \left(1 - \frac{F_{k+2}}{2^k}\right) + \frac{F_k}{2^{k+1}} \qquad \text{(using (1))}$$

$$= 1 - \frac{1}{2^k}\left[F_{k+2} - \frac{F_k}{2}\right] = 1 - \frac{1}{2^{k+1}}\left[F_{k+2} + F_{k+2} - F_k\right].$$

But $F_{k+2} = F_{k+1} + F_k \Rightarrow F_{k+2} - F_k = F_{k+1}$.

$$\therefore \sum_{i=1}^{k+1} \frac{F_{i-1}}{2^i} = 1 - \frac{1}{2^{k+1}}\left[F_{k+2} + F_{k+1}\right] = 1 - \frac{F_{k+3}}{2^{k+1}} \qquad (\because F_{k+3} = F_{k+2} + F_{k+1}).$$

This is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

**Problem 10.** QP If $F_i$'s are Fibonacci numbers ($F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$) and $L_i$'s are Lucas numbers defined recursively as $L_0 = 2,\ L_1 = 1$ and $L_n = L_{n-1} + L_{n-2}\ \forall n \geq 2$, prove that $L_n = F_{n-1} + F_{n+1}\ \forall n \geq 1$.

**Solution.** Let $S(n): \ L_n = F_{n-1} + F_{n+1}\ \forall n \geq 1$.

Basic step: $S(1): \ L_1 = F_0 + F_2$, i.e., $1 = 0 + 1 = 1$ ($\because F_2 = F_1 + F_0 = 1$). Also $S(2): \ L_2 = L_1 + L_0 = 3$ and $F_1 + F_3 = 1 + 2 = 3$. $\therefore S(n)$ is true for $n = 1, 2$.

Induction step: Assume $S(k-1)$ and $S(k)$ are true for $k \geq 2$: $L_k = F_{k-1} + F_{k+1},\ L_{k-1} = F_{k-2} + F_k$. To prove: $L_{k+1} = F_k + F_{k+2}$. Consider

$$\begin{aligned} L_{k+1} &= L_k + L_{k-1} \qquad \text{(by the definition of } L_n) \\ &= (F_{k-1} + F_{k+1}) + (F_{k-2} + F_k) \qquad \text{(by the assumption made)} \\ &= (F_{k-1} + F_{k-2}) + (F_{k+1} + F_k) = F_k + F_{k+2} \qquad \text{(by the definition of } F_n). \end{aligned}$$

This is the statement $S(k+1)$. Thus, by mathematical induction, the required result is true $\forall n \geq 1$. ■

**Problem 11.** The Ackermann's numbers are defined recursively for $m, n \in \mathbb{N}$ as follows:

$$A_{0,n} = n + 1 \ \text{ for } n \geq 0 \tag{1}$$
$$A_{m,0} = A_{m-1,1} \ \text{ for } m > 0 \tag{2}$$
$$A_{m,n} = A_{m-1,p}, \ \text{where } p = A_{m,n-1} \ \text{ for } m, n > 0 \tag{3}$$

Prove that $A_{1,n} = n + 2\ \forall n \in \mathbb{N}$.

**Solution.** Let $S(n): \ A_{1,n} = n + 2$.

Basic step: $S(0)$: by (2) and (1), $A_{1,0} = A_{0,1} = 1 + 1 = 2 = 0 + 2$. True.
$S(1): \ A_{1,1} = 1 + 2 \Rightarrow$ RHS $= 3$. LHS: $A_{1,1} = A_{0,p}$, where $p = A_{1,0} = A_{0,1} = 2$ (by (2) and (1)). $\therefore A_{1,1} = A_{0,2} = 2 + 1 = 3$ (by (1)). LHS = RHS. $\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true for $k \geq 1$, i.e., $A_{1,k} = k + 2$. (4)
To prove: $A_{1,k+1} = k + 3$. Consider

$$\begin{aligned} A_{1,k+1} &= A_{0,p} \ \text{(using (3))}, \quad \text{where } p = A_{1,k} = k + 2 \ \text{(using (4))} \\ &= A_{0,k+2} = k + 2 + 1 = k + 3 \qquad \text{(using (1))}, \end{aligned}$$

which is the statement $S(k+1)$. Hence, by mathematical induction, $A_{1,n} = n + 2\ \forall n \in \mathbb{N}$. ■

## Practice Questions — Recursive Definitions

1. Give a recursive definition of (i) $a_n = 3n + 2$ (ii) $a_n = n^2$ (iii) $a_n = 5 \cdot 2^n$ (iv) $a_n = 1 + (-1)^n$.
2. Find the explicit formula for $a_1 = 3,\ a_n = a_{n-1} + 4\ (n \geq 2)$.
3. Find the explicit formula for $a_0 = 2,\ a_n = 3a_{n-1}\ (n \geq 1)$.
4. Prove that $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$ for all $n \geq 1$.
5. Prove that $F_1 + F_3 + F_5 + \cdots + F_{2n-1} = F_{2n}$ for all $n \geq 1$.
6. A sequence is defined by $a_1 = 1,\ a_2 = 3,\ a_n = a_{n-1} + a_{n-2}$. Prove that $a_n < \left(\frac{7}{4}\right)^n$ for all $n \geq 1$. QP
7. Using the Ackermann definition above, prove that $A_{2,n} = 2n + 3$ for all $n \in \mathbb{N}$.
8. Give a recursive definition of the set of all positive odd integers.

PART B
# Properties of Integers NEW

## 3 The Division Algorithm

**Divisibility**

Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say $a$ **divides** $b$, written $a \mid b$, if there is an integer $c$ such that $b = ac$. Then $a$ is a *divisor* (factor) of $b$ and $b$ is a *multiple* of $a$. If $a$ does not divide $b$ we write $a \nmid b$.
*Examples:* $3 \mid 12,\ 7 \mid (-21),\ 5 \mid 0,\ 4 \nmid 10$.

**Properties of divisibility**

For integers $a, b, c$ (with $a \neq 0$ where it is a divisor):

(i) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

(ii) If $a \mid b$, then $a \mid bc$ for every integer $c$.

(iii) If $a \mid b$ and $b \mid c$, then $a \mid c$ (transitivity).

(iv) If $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for all integers $m, n$ (linear combination).

**Proof.** of (iv): $b = as$, $c = at$ for some $s, t \in \mathbb{Z}$. Then $mb + nc = a(ms + nt)$, and $ms + nt \in \mathbb{Z}$. So $a \mid (mb + nc)$. The others are proved the same way. ■

**The Division Algorithm** QP

Let $a$ be an integer and $b$ a positive integer. Then there exist **unique** integers $q$ and $r$ such that

$$a = bq + r, \qquad 0 \leq r < b.$$

$a$ is the *dividend*, $b$ the *divisor*, $q$ the *quotient* and $r$ the *remainder*. We write

$$q = a \ \mathbf{div}\ b = \left\lfloor \tfrac{a}{b} \right\rfloor, \qquad r = a \bmod b.$$

**Proof.** Existence (using the Well-Ordering Principle). Consider

$$S = \{\, a - bk \ :\ k \in \mathbb{Z},\ a - bk \geq 0 \,\}.$$

$S$ is non-empty: take $k = -|a|$; then $a - bk = a + b|a| \geq a + |a| \geq 0$. If $0 \in S$, take $r = 0$. Otherwise $S$ is a non-empty set of positive integers, so by the Well-Ordering Principle it has a least element $r = a - bq$ for some $q$. Clearly $r \geq 0$. If $r \geq b$, then $r - b = a - b(q + 1) \geq 0$ is an element of $S$ smaller than $r$ — a contradiction. Hence $0 \leq r < b$ and $a = bq + r$.

Uniqueness. Suppose $a = bq_1 + r_1 = bq_2 + r_2$ with $0 \leq r_1, r_2 < b$. Then $b(q_1 - q_2) = r_2 - r_1$, so $b \mid (r_2 - r_1)$. But $-b < r_2 - r_1 < b$, and the only multiple of $b$ in this range is 0. So $r_1 = r_2$, and then $q_1 = q_2$. ■

**Note:**

1. $b \mid a$ if and only if $a \bmod b = 0$.
2. The remainder is *never negative*. For negative $a$, choose $q$ one less than the "calculator" quotient: $-11 = 3(-4) + 1$, *not* $3(-3) - 2$.
3. In C/Java the operator `%` may return a negative value for negative operands ( $-11 \,\%\, 3 = -2$); the mathematical $\bmod$ always returns $0 \leq r < b$.

### 3.1 Examples

**Example 1.** Find the quotient and remainder when (i) 101 is divided by 11; (ii) $-11$ is divided by 3; (iii) $-45$ is divided by 7; (iv) 7 is divided by 15.

**Solution.** (i) $101 = 11 \times 9 + 2$: $q = 9,\ r = 2$. (ii) $-11 = 3 \times (-4) + 1$: $q = -4,\ r = 1$.
(iii) $-45 = 7 \times (-7) + 4$: $q = -7,\ r = 4$. (iv) $7 = 15 \times 0 + 7$: $q = 0,\ r = 7$.

**Example 2.** Today is Sunday. What day of the week will it be after 100 days?

**Solution.** Days repeat in cycles of 7. $100 = 7 \times 14 + 2$, i.e., $100 \bmod 7 = 2$. After 14 complete weeks it is Sunday again; 2 more days give **Tuesday**.

**Example 3.** Show that the square of any integer is of the form $4k$ or $4k + 1$.

**Solution.** By the division algorithm with $b = 2$, every integer $n$ is $2m$ or $2m + 1$.

- If $n = 2m$: $n^2 = 4m^2 = 4k$ with $k = m^2$.
- If $n = 2m + 1$: $n^2 = 4m^2 + 4m + 1 = 4(m^2 + m) + 1 = 4k + 1$.

Hence $n^2$ is of the form $4k$ or $4k + 1$. (Consequently, a number like 4003, which is of the form $4k + 3$, is never a perfect square.) ■

**Example 4.** Prove that if $n$ is an odd integer, then $8 \mid (n^2 - 1)$.

**Solution.** Let $n = 2m + 1$. Then $n^2 - 1 = 4m^2 + 4m = 4m(m + 1)$. Among two consecutive integers $m, m + 1$, one is even, so $m(m + 1) = 2t$. Hence $n^2 - 1 = 8t$, i.e., $8 \mid (n^2 - 1)$. ■

**Example 5.** Prove that the product of any three consecutive integers is divisible by 6.

**Solution.** Let the integers be $n,\ n + 1,\ n + 2$. By the division algorithm, $n = 3q,\ 3q + 1$ or $3q + 2$; in each case exactly one of the three numbers is a multiple of 3 (respectively $n,\ n + 2,\ n + 1$). Also at least one of them is even. Since $2 \mid P$ and $3 \mid P$ with $\gcd(2, 3) = 1$, we get $6 \mid P$, where $P = n(n + 1)(n + 2)$. ■

**Example 6.** If $a \mid b$ and $a \mid c$, prove that $a \mid (3b - 5c)$. Hence show that $7 \mid (3 \cdot 91 - 5 \cdot 35)$.

**Solution.** By property (iv) with $m = 3,\ n = -5$. Since $7 \mid 91$ and $7 \mid 35$, $7 \mid (273 - 175) = 98$. Check: $98 = 7 \times 14$. ■

**Practice Questions — Division Algorithm**

1. Find $q$ and $r$ when (i) 789 is divided by 23; (ii) $-100$ is divided by 13; (iii) 0 is divided by 19; (iv) $-1$ is divided by 3.
2. Evaluate: (i) $-17 \bmod 5$ (ii) $144 \bmod 7$ (iii) $-101$ **div** 13.
3. Show that the cube of any integer is of the form $9k$, $9k + 1$ or $9k - 1$.
4. Show that every odd integer is of the form $4k + 1$ or $4k + 3$.
5. Prove that $3 \mid (n^3 - n)$ for every integer $n$ using the division algorithm (compare with the induction proof in Part A).
6. If $a \mid b$ and $b \mid a$ for positive integers $a, b$, prove that $a = b$.
7. A clock shows 9 o'clock. What will it show after 100 hours?

## 4 Prime Numbers

**Prime and composite numbers**

An integer $p > 1$ is called **prime** if its only positive divisors are 1 and $p$. An integer $n > 1$ that is not prime is called **composite**; i.e., $n = ab$ for some integers $1 < a, b < n$.
*Note:* 1 is neither prime nor composite. 2 is the only even prime.

**Theorem 1 (Test for primality)** QP

If $n$ is a composite integer, then $n$ has a prime divisor less than or equal to $\sqrt{n}$.
*Consequently:* to test whether $n$ is prime, it is enough to try dividing $n$ by the primes $p \le \sqrt{n}$.

**Proof.** Since $n$ is composite, $n = ab$ with $1 < a \le b < n$. If $a > \sqrt{n}$, then $b \ge a > \sqrt{n}$ and $ab > n$, a contradiction. So $a \le \sqrt{n}$. By Example 1 of Section 1.2, $a$ has a prime divisor $p$, and $p \le a \le \sqrt{n}$. Since $p \mid a$ and $a \mid n$, $p \mid n$. ■

**Theorem 2 (Euclid)** QP

There are infinitely many primes.

**Proof.** (By contradiction.) Suppose there are only finitely many primes $p_1, p_2, \ldots, p_n$. Let

$$Q = p_1 p_2 \cdots p_n + 1.$$

Since $Q > 1$, $Q$ has a prime divisor $p$, which must be one of the $p_i$. Then $p \mid p_1 p_2 \cdots p_n$ and $p \mid Q$, so $p \mid \big(Q - p_1 p_2 \cdots p_n\big) = 1$, which is impossible. Hence there are infinitely many primes. ■

> **Caution:** $Q$ need not itself be prime. E.g. $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$. The proof only needs a prime divisor of $Q$ that is not in the list.

### The Sieve of Eratosthenes

To list all primes up to $n$: write $2, 3, \ldots, n$; circle 2 and strike out its multiples; circle the next unstruck number (3) and strike out its multiples; continue until the circled prime exceeds $\sqrt{n}$. The remaining numbers are prime. For $n = 50$ we only sieve with $2, 3, 5, 7$ (since $\sqrt{50} < 8$):

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **2** | **3** | 4 | **5** | 6 | **7** | 8 | 9 | 10 |
| **11** | 12 | **13** | 14 | 15 | 16 | **17** | 18 | **19** | 20 |
| 21 | 22 | **23** | 24 | 25 | 26 | 27 | 28 | **29** | 30 |
| **31** | 32 | 33 | 34 | 35 | 36 | **37** | 38 | 39 | 40 |
| **41** | 42 | **43** | 44 | 45 | 46 | **47** | 48 | 49 | 50 |

There are **15 primes** up to 50 (shown in bold).

## 4.1 Examples

**Example 1.** Determine whether (i) 401 and (ii) 391 are prime.

**Solution.** (i) $\sqrt{401} \approx 20.02$; primes $\le 20$ are 2, 3, 5, 7, 11, 13, 17, 19. None divides 401 (remainders 1, 2, 1, 2, 5, 11, 10, 2). Hence **401 is prime**.
(ii) $\sqrt{391} \approx 19.8$. Trying the primes in turn: $391 = 17 \times 23$. Hence **391 is composite**.

**Example 2.** Show that the formula $n^2 + n + 41$ does not always give a prime.

**Solution.** It gives primes for $n = 0, 1, \ldots, 39$, but for $n = 40$: $40^2 + 40 + 41 = 1600 + 80 + 41 = 1681 = 41^2$, which is composite. (A pattern that holds for many cases is not a proof!)

**Example 3.** Prove that every prime $p > 3$ is of the form $6k + 1$ or $6k - 1$.

**Solution.** By the division algorithm, $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$. If $r = 0, 2, 4$ then $2 \mid p$; if $r = 3$ then $3 \mid p$. Since $p > 3$ is prime, neither is possible. So $r = 1$ or $r = 5$, i.e., $p = 6k + 1$ or $p = 6k + 5 = 6(k + 1) - 1$. ■

**Example 4.** Prove that if $2^n - 1$ is prime, then $n$ is prime. Is the converse true? Test $2^{11} - 1$.

**Solution.** (Contrapositive.) If $n = ab$ with $1 < a, b < n$, then using $x^b - 1 = (x-1)(x^{b-1} + \cdots + x + 1)$ with $x = 2^a$:

$$2^{ab} - 1 = (2^a - 1)\big(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1\big),$$

and $1 < 2^a - 1 < 2^n - 1$, so $2^n - 1$ is composite. Hence $2^n - 1$ prime $\Rightarrow$ $n$ prime.
The converse is false: 11 is prime but $2^{11} - 1 = 2047 = 23 \times 89$. ■

**Practice Questions — Prime Numbers**

1. Determine which of the following are prime: 221, 437, 509, 1001.
2. List all primes between 50 and 100.
3. Use the Sieve of Eratosthenes to find all primes up to 100. How many are there?
4. Prove that if $p$ and $p^2 + 2$ are both prime, then $p = 3$. (Hint: consider $p \bmod 3$.)
5. List the first eight primes of the form $4k + 3$.
6. Find five consecutive composite integers. (Hint: $6! + 2,\ 6! + 3,\ \ldots$)

# 5 Greatest Common Divisor and the Euclidean Algorithm

**GCD, LCM and relatively prime integers**

Let $a, b$ be integers, not both zero. The largest integer $d$ such that $d \mid a$ and $d \mid b$ is the **greatest common divisor** of $a$ and $b$, written $\gcd(a, b)$.
$a$ and $b$ are **relatively prime** (co-prime) if $\gcd(a, b) = 1$. Integers $a_1, a_2, \ldots, a_n$ are *pairwise relatively prime* if $\gcd(a_i, a_j) = 1$ for all $i \neq j$.
The **least common multiple** $\operatorname{lcm}(a, b)$ of positive integers $a, b$ is the smallest positive integer divisible by both $a$ and $b$.
*Examples:* $\gcd(24, 36) = 12$; $\gcd(17, 22) = 1$; $\operatorname{lcm}(4, 6) = 12$; 10, 17, 21 are pairwise relatively prime.

**Lemma (basis of the Euclidean Algorithm)** QP

If $a = bq + r$, where $a, b, q, r$ are integers, then $\gcd(a, b) = \gcd(b, r)$.

**Proof.** If $d \mid a$ and $d \mid b$, then $d \mid (a - bq) = r$. So every common divisor of $a, b$ is a common divisor of $b, r$. Conversely, if $d \mid b$ and $d \mid r$, then $d \mid (bq + r) = a$. So $a, b$ and $b, r$ have exactly the same common divisors, and hence the same greatest common divisor. ■

### The Euclidean Algorithm

Let $a \geq b > 0$. Apply the division algorithm repeatedly:

$$\begin{aligned}
a &= bq_1 + r_1, & 0 < r_1 < b \\
b &= r_1 q_2 + r_2, & 0 < r_2 < r_1 \\
r_1 &= r_2 q_3 + r_3, & 0 < r_3 < r_2 \\
&\vdots \\
r_{n-2} &= r_{n-1} q_n + r_n, & 0 < r_n < r_{n-1} \\
r_{n-1} &= r_n q_{n+1} + 0.
\end{aligned}$$

The remainders strictly decrease, so a zero remainder is eventually reached. By the Lemma, $\gcd(a, b) = \gcd(b, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n$.

$\gcd(a, b)$ = the **last non-zero remainder**.

**Algorithm: Euclid (pseudocode)**

**procedure** $\gcd(a, b$: positive integers)
  $x := a;\ y := b$
  **while** $y \neq 0$
    $r := x \bmod y$
    $x := y;\ y := r$
  **return** $x$  $\{x = \gcd(a, b)\}$

Recursive form: $\gcd(a, 0) = a; \quad \gcd(a, b) = \gcd(b,\ a \bmod b)$.

**Bézout's Identity** QP

If $a$ and $b$ are positive integers, then there exist integers $s$ and $t$ such that

$$\gcd(a, b) = sa + tb.$$

($s, t$ are called Bézout coefficients; they are found by running the Euclidean algorithm *backwards* — the **extended Euclidean algorithm**.)

**Consequences**

(i) **Euclid's Lemma.** If $p$ is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally, if $p \mid a_1 a_2 \cdots a_n$ then $p \mid a_i$ for some $i$.

(ii) If $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

(iii) For positive integers $a, b$: $\gcd(a, b) \times \operatorname{lcm}(a, b) = a \times b$.

**Proof.** of (ii) (and hence (i)): Since $\gcd(a, b) = 1$, by Bézout $1 = sa + tb$. Multiplying by $c$: $c = sac + tbc$. Now $a \mid sac$ and $a \mid bc$ (given), so $a \mid (sac + tbc) = c$. For (i): if $p \nmid a$, then $\gcd(p, a) = 1$ (the only positive divisors of $p$ are 1 and $p$), and by (ii) $p \mid b$. ■

## 5.1 Examples

**Example 1.** QP Find $\gcd(414, 662)$ using the Euclidean algorithm.

**Solution.**

$$\begin{aligned} 662 &= 414 \times 1 + 248 \\ 414 &= 248 \times 1 + 166 \\ 248 &= 166 \times 1 + 82 \\ 166 &= 82 \times 2 + 2 \\ 82 &= 2 \times 41 + 0 \end{aligned}$$

The last non-zero remainder is 2. $\therefore \gcd(414, 662) = 2$.

**Example 2.** QP Find $\gcd(1071, 462)$ and express it in the form $1071s + 462t$.

**Solution.** *Forward (Euclid):*

$$\begin{aligned} 1071 &= 462 \times 2 + 147 && \Rightarrow\ 147 = 1071 - 2 \times 462 \\ 462 &= 147 \times 3 + 21 && \Rightarrow\ 21 = 462 - 3 \times 147 \\ 147 &= 21 \times 7 + 0 \end{aligned}$$

$\gcd(1071, 462) = 21$.

*Backward (extended Euclid):*

$$\begin{aligned} 21 &= 462 - 3 \times 147 = 462 - 3\,(1071 - 2 \times 462) \\ &= 462 - 3 \times 1071 + 6 \times 462 = 7 \times 462 - 3 \times 1071. \end{aligned}$$

$\therefore 21 = (-3) \times 1071 + 7 \times 462$, i.e., $s = -3,\ t = 7$. Check: $-3213 + 3234 = 21$. ✓

**Example 3.** Find $\gcd(252, 198)$, express it as a linear combination of 252 and 198, and find $\operatorname{lcm}(252, 198)$.

**Solution.**

$$\begin{aligned} 252 &= 198 \times 1 + 54 \\ 198 &= 54 \times 3 + 36 \\ 54 &= 36 \times 1 + 18 \\ 36 &= 18 \times 2 + 0 \end{aligned} \qquad \begin{aligned} 54 &= 252 - 198 \\ 36 &= 198 - 3 \times 54 \\ 18 &= 54 - 36 \end{aligned}$$

$\gcd = 18$. Back-substituting:

$$\begin{aligned} 18 &= 54 - 36 = 54 - (198 - 3 \times 54) = 4 \times 54 - 198 \\ &= 4\,(252 - 198) - 198 = 4 \times 252 - 5 \times 198. \end{aligned}$$

Check: $1008 - 990 = 18$ ✓. $\operatorname{lcm}(252, 198) = \dfrac{252 \times 198}{18} = \dfrac{49896}{18} = 2772.$

**Example 4.** Prove that any two consecutive integers are relatively prime.

**Solution.** Let $d = \gcd(n, n+1)$. Then $d \mid n$ and $d \mid (n+1)$, so $d \mid \big((n+1) - n\big) = 1$. Hence $d = 1$. ■

**Example 5.** If $\gcd(a, b) = d$, prove that $\gcd\big(\frac{a}{d}, \frac{b}{d}\big) = 1$.

**Solution.** By Bézout, $d = sa + tb$. Dividing by $d$: $1 = s\frac{a}{d} + t\frac{b}{d}$. Any common divisor of $\frac{a}{d}$ and $\frac{b}{d}$ divides the right-hand side, hence divides 1. So the gcd is 1. ■

**Example 6.** (*Counting the work.*) How many divisions does the Euclidean algorithm use to find $\gcd(89, 55)$?

**Solution.** $89 = 55 \cdot 1 + 34,\ 55 = 34 \cdot 1 + 21,\ 34 = 21 \cdot 1 + 13,\ 21 = 13 \cdot 1 + 8,\ 13 = 8 \cdot 1 + 5,\ 8 = 5 \cdot 1 + 3,\ 5 = 3 \cdot 1 + 2,\ 3 = 2 \cdot 1 + 1,\ 2 = 1 \cdot 2 + 0.$
9 divisions, $\gcd = 1$. Consecutive Fibonacci numbers are the *worst case* for Euclid (Lamé's theorem): the number of divisions is $O(\log b)$, which is why the algorithm is fast even for 300-digit numbers used in RSA.

**Practice Questions — GCD and Euclidean Algorithm**

1. Use the Euclidean algorithm to find (i) $\gcd(1529, 14039)$ (ii) $\gcd(1001, 1331)$ (iii) $\gcd(2740, 1760)$. QP
2. Express $\gcd(56, 72)$ as a linear combination of 56 and 72.
3. Express $\gcd(123, 277)$ in the form $123s + 277t$.
4. Find $\operatorname{lcm}(1001, 1331)$ using $\gcd \times \operatorname{lcm} = ab$.
5. Prove that $\gcd(a, b) = \gcd(a - b,\ b)$ for integers $a > b > 0$.
6. Prove that if $\gcd(a, m) = \gcd(b, m) = 1$, then $\gcd(ab, m) = 1$.
7. Write a recursive program (C/Python) for $\gcd(a, b)$ and trace it for $(1071, 462)$.

# 6 Modular Arithmetic

**Congruence modulo $m$** QP

Let $m$ be a positive integer. Integers $a$ and $b$ are **congruent modulo** $m$, written

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. Equivalently, $a \bmod m = b \bmod m$ (same remainder on division by $m$). $m$ is the *modulus*.
*Examples:* $17 \equiv 5 \pmod 6$ since $6 \mid 12$; $-8 \equiv 7 \pmod 5$ since $5 \mid (-15)$; $24 \not\equiv 14 \pmod 6$ since $6 \nmid 10$.

**Theorem: Congruence is an equivalence relation**

For a fixed $m$, the relation $a \equiv b \pmod m$ on $\mathbb{Z}$ is

(i) *reflexive*: $a \equiv a$, since $m \mid 0$;

(ii) *symmetric*: $a \equiv b \Rightarrow b \equiv a$, since $m \mid (a-b) \Rightarrow m \mid (b-a)$;

(iii) *transitive*: $a \equiv b,\ b \equiv c \Rightarrow a \equiv c$, since $a - c = (a-b) + (b-c)$.

Its equivalence classes are the $m$ **residue classes** $[0], [1], \ldots, [m-1]$ (link to Unit 1, Equivalence Relations).

**Theorem: Arithmetic with congruences** QP

If $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd, \qquad a^k \equiv b^k \pmod m \ (k \geq 1).$$

Hence $(a+b) \bmod m = \big((a \bmod m) + (b \bmod m)\big) \bmod m$ and $(ab) \bmod m = \big((a \bmod m)(b \bmod m)\big) \bmod m$.

**Proof.** $a = b + sm$, $c = d + tm$. Then $a + c = (b+d) + (s+t)m$ and $ac = bd + (bt + ds + stm)m$. The power rule follows by induction on $k$. ■

**Caution:** Division is *not* always allowed. $14 \equiv 8 \pmod 6$, but dividing by 2 gives $7 \not\equiv 4 \pmod 6$. We may cancel $c$ from $ac \equiv bc \pmod m$ only when $\gcd(c, m) = 1$.

**Arithmetic modulo $m$ (the set $\mathbb{Z}_m$)**

$\mathbb{Z}_m = \{0, 1, 2, \ldots, m-1\}$ with operations $a +_m b = (a+b) \bmod m$ and $a \cdot_m b = (ab) \bmod m$. E.g. in $\mathbb{Z}_{11}$: $7 +_{11} 9 = 5$ and $7 \cdot_{11} 9 = 8$.

**Modular inverse and linear congruences**

An integer $\bar{a}$ with $a\bar{a} \equiv 1 \pmod m$ is an **inverse of $a$ modulo $m$**. It *exists if and only if* $\gcd(a, m) = 1$, and is found from Bézout: if $1 = sa + tm$, then $sa \equiv 1 \pmod m$, so $\bar{a} = s$.
The **linear congruence** $ax \equiv b \pmod m$ with $\gcd(a, m) = 1$ has the solution $x \equiv \bar{a}\, b \pmod m$.

## 6.1 Examples

**Example 1.** QP Find the remainder when $2^{100}$ is divided by 7.

**Solution.** $2^3 = 8 \equiv 1 \pmod 7$. So $2^{99} = (2^3)^{33} \equiv 1^{33} = 1$. Hence $2^{100} = 2^{99} \cdot 2 \equiv 2 \pmod 7$. Remainder $= \mathbf{2}$.

**Example 2.** Find the last digit of $7^{222}$.

**Solution.** The last digit is $7^{222} \bmod 10$. Powers of 7 mod 10: $7, 9, 3, 1, 7, 9, \ldots$ (cycle of length 4, since $7^4 = 2401 \equiv 1$). $222 = 4 \times 55 + 2$, so $7^{222} = (7^4)^{55} \cdot 7^2 \equiv 1 \cdot 49 \equiv 9 \pmod{10}$. Last digit $= \mathbf{9}$.

**Example 3.** Using congruences, prove that $7 \mid \left(3^{2n+1} + 2^{n+2}\right)$ for all $n \geq 0$.

**Solution.** $3^{2n+1} + 2^{n+2} = 3 \cdot 9^n + 4 \cdot 2^n$. Since $9 \equiv 2 \pmod 7$, $9^n \equiv 2^n$. Hence

$$3 \cdot 9^n + 4 \cdot 2^n \equiv 3 \cdot 2^n + 4 \cdot 2^n = 7 \cdot 2^n \equiv 0 \pmod 7.$$

(Compare with the induction proof of Problem 10 in Section 1 — congruences often give shorter proofs.) ■

**Example 4.** Find an inverse of 3 modulo 7 and solve $3x \equiv 4 \pmod 7$.

**Solution.** $\gcd(3, 7) = 1$. $7 = 3 \cdot 2 + 1 \Rightarrow 1 = 7 - 2 \cdot 3$, so $(-2) \cdot 3 \equiv 1 \pmod 7$, i.e., $\bar{3} \equiv -2 \equiv 5$. (Check: $3 \cdot 5 = 15 \equiv 1$.)
$x \equiv 5 \cdot 4 = 20 \equiv 6 \pmod 7$. Check: $3 \cdot 6 = 18 \equiv 4 \pmod 7$ ✓. All solutions: $x = 6 + 7k,\ k \in \mathbb{Z}$.

**Example 5.** Find the inverse of 101 modulo 4620.

**Solution.** Euclid: $4620 = 101 \cdot 45 + 75$; $101 = 75 \cdot 1 + 26$; $75 = 26 \cdot 2 + 23$; $26 = 23 \cdot 1 + 3$; $23 = 3 \cdot 7 + 2$; $3 = 2 \cdot 1 + 1$. So $\gcd = 1$. Back-substitution gives

$$1 = 1601 \cdot 101 - 35 \cdot 4620.$$

Hence $\overline{101} \equiv \mathbf{1601} \pmod{4620}$. (Check: $101 \times 1601 = 161701 = 35 \times 4620 + 1$ ✓.)

**Example 6.** (*Application: hashing.*) A college stores student records in 111 memory locations using $h(k) = k \bmod 111$, where $k$ is the 9-digit ID. Find the locations for IDs 064212848 and 037149212.

**Solution.** $064212848 \bmod 111 = 14$ and $037149212 \bmod 111 = 65$. So the records go to locations 14 and 65. (If a location is occupied — a *collision* — the next free location is used.)

**Practice Questions — Modular Arithmetic**

1. Decide whether each is true: (i) $80 \equiv 5 \pmod{17}$ (ii) $103 \equiv 5 \pmod{17}$ (iii) $-29 \equiv 13 \pmod{7}$ (iv) $-122 \equiv -22 \pmod{10}$.
2. Find the remainder when (i) $3^{100}$ is divided by 5; (ii) $5^{40}$ is divided by 11; (iii) $2^{50} + 3^{50}$ is divided by 13.
3. Find the last two digits of $3^{20}$.
4. Find inverses of (i) 4 mod 9 (ii) 19 mod 141 (iii) 55 mod 89.
5. Solve: (i) $4x \equiv 5 \pmod{9}$ (ii) $19x \equiv 4 \pmod{141}$.
6. Using congruences, prove that $43 \mid (6^{n+2} + 7^{2n+1})$ for all $n \geq 1$. (Hint: $7^2 \equiv 6 \pmod{43}$.)
7. Show that $a \equiv b \pmod{m}$ implies $\gcd(a, m) = \gcd(b, m)$.

# 7 The Fundamental Theorem of Arithmetic

**The Fundamental Theorem of Arithmetic** QP

Every integer $n > 1$ can be written as a product of primes, and this factorisation is **unique** apart from the order of the factors. Writing equal primes together, every $n > 1$ has a unique *canonical (standard) form*

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}, \qquad p_1 < p_2 < \cdots < p_k \text{ primes}, \ e_i \geq 1.$$

**Proof.** Existence (strong induction). $S(n)$: $n$ is a product of primes. $S(2)$ is true since 2 is prime. Assume $S(2), \ldots, S(k)$ are true. If $k+1$ is prime, it is a product of one prime. Otherwise $k+1 = ab$ with $2 \leq a, b \leq k$; by hypothesis $a$ and $b$ are products of primes, and so is $ab = k+1$. Hence $S(n)$ holds $\forall n \geq 2$.

Uniqueness. Suppose $n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$ (primes, not necessarily distinct). Then $p_1 \mid q_1 q_2 \cdots q_s$, and by Euclid's Lemma $p_1 \mid q_j$ for some $j$. Since $q_j$ is prime, $p_1 = q_j$. Cancel this common factor and repeat. The factors pair off exactly, so $r = s$ and the two lists are the same up to order. ■

**Why 1 is not called a prime:** if it were, $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3$ and uniqueness would fail.

**Applications of the canonical form**

Let $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$.

(i) Every positive divisor of $n$ is $d = p_1^{f_1} \cdots p_k^{f_k}$ with $0 \leq f_i \leq e_i$. Hence (by the product rule)

$$\tau(n) = \text{number of divisors} = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1).$$

(ii) Sum of divisors: $\sigma(n) = \displaystyle\prod_{i=1}^{k} \frac{p_i^{e_i+1} - 1}{p_i - 1}$.

(iii) If $a = \prod p_i^{a_i}$ and $b = \prod p_i^{b_i}$ (allowing exponent 0), then

$$\gcd(a, b) = \prod p_i^{\min(a_i, b_i)}, \qquad \operatorname{lcm}(a, b) = \prod p_i^{\max(a_i, b_i)}.$$

Since $\min + \max = a_i + b_i$, this proves $\gcd(a,b) \cdot \text{lcm}(a,b) = ab$.

## 7.1 Examples

**Example 1.** Find the prime factorisations of 100, 641, 999, 1024 and 360.

**Solution.** $100 = 2^2 \cdot 5^2$; 641 is prime (no prime $\leq 25$ divides it); $999 = 3^3 \cdot 37$; $1024 = 2^{10}$; $360 = 2^3 \cdot 3^2 \cdot 5$.

**Example 2.** QP Find the number of positive divisors of 360 and their sum.

**Solution.** $360 = 2^3 \cdot 3^2 \cdot 5^1$. $\tau(360) = (3+1)(2+1)(1+1) = 24$.

$$\sigma(360) = \frac{2^4-1}{1} \cdot \frac{3^3-1}{2} \cdot \frac{5^2-1}{4} = 15 \times 13 \times 6 = 1170.$$

(Compare: Problem 5 in Section 8 counts the divisors of 441000 in the same way.)

**Example 3.** Find $\gcd(120, 500)$ and $\text{lcm}(120, 500)$ using prime factorisation.

**Solution.** $120 = 2^3 \cdot 3^1 \cdot 5^1$ and $500 = 2^2 \cdot 3^0 \cdot 5^3$.
$\gcd = 2^2 \cdot 3^0 \cdot 5^1 = 20$; $\text{lcm} = 2^3 \cdot 3^1 \cdot 5^3 = 3000$. Check: $20 \times 3000 = 60000 = 120 \times 500$ ✓.

**Example 4.** QP Prove that $\sqrt{2}$ is irrational.

**Solution.** Suppose $\sqrt{2} = \frac{a}{b}$ with $a, b \in \mathbb{Z}^+$. Then $a^2 = 2b^2$. In the prime factorisation of $a^2$, the exponent of 2 is *even* (twice the exponent in $a$). In $2b^2$, the exponent of 2 is *odd* (one more than an even number). By the uniqueness of factorisation, $a^2 \neq 2b^2$ — a contradiction. Hence $\sqrt{2}$ is irrational. ■

**Example 5.** How many zeros are there at the end of 100!?

**Solution.** Each trailing zero comes from a factor $10 = 2 \times 5$. In 100! there are more 2's than 5's, so we count the exponent of 5:

$$\left\lfloor \tfrac{100}{5} \right\rfloor + \left\lfloor \tfrac{100}{25} \right\rfloor = 20 + 4 = 24.$$

100! ends in **24 zeros**.

**Example 6.** Find the smallest positive integer with exactly 12 divisors.

**Solution.** $\tau(n) = 12$ can arise as $12,\ 6 \times 2,\ 4 \times 3,\ 3 \times 2 \times 2$, i.e., $n = p^{11},\ p^5 q,\ p^3 q^2,\ p^2 qr$. Using the smallest primes: $2^{11} = 2048$, $2^5 \cdot 3 = 96$, $2^3 \cdot 3^2 = 72$, $2^2 \cdot 3 \cdot 5 = 60$. The smallest is **60**.

**Practice Questions — Fundamental Theorem of Arithmetic**

1. Find the prime factorisation of (i) 1001 (ii) 7007 (iii) 1023 (iv) 10!.
2. Find the number and the sum of the positive divisors of 720.
3. Find the number of proper divisors of 10800. QP
4. Find gcd and lcm of $2^3 \cdot 3^5 \cdot 5^2$ and $2^5 \cdot 3^3 \cdot 7$.
5. Prove that $\sqrt{3}$ is irrational. Is $\sqrt{4}$? Where does your proof break?
6. How many zeros are there at the end of 1000!?
7. Prove that $n$ is a perfect square if and only if $\tau(n)$ is odd.

PART C

# Basic Counting Techniques

## 8 The Rules of Sum and Product

**The Sum Rule**

If $T_1$ and $T_2$ are two tasks such that $T_1$ can be performed in $m$ different ways, $T_2$ can be performed in $n$ different ways, and these two tasks cannot be performed simultaneously, then one of the tasks ($T_1$ or $T_2$) can be performed in $m + n$ ways.
*Example:* Suppose a library has 12 books on Mathematics, 10 books on Physics and 16 books on Computers. If a student wishes to choose one of these books to study, then the number of ways he can choose the book is $12 + 10 + 16 = 38$.

**The Product Rule**

If $T_1$ and $T_2$ are two tasks such that $T_1$ can be performed in $m$ different ways and, for each of these ways, $T_2$ can be performed in $n$ different ways, then both the tasks can be performed in $m \times n$ different ways.
*Example:* Suppose a person has 8 shirts and 5 ties. Then the number of ways of choosing a shirt and a tie is $8 \times 5 = 40$.

### 8.1 Problems

**Problem 1.** A licence plate consists of two English letters followed by 4 digits. If repetitions are allowed, how many of the plates have only vowels and even digits?

**Solution.** Each of the first 2 positions can be filled in 5 ways (with the 5 vowels), and each of the remaining 4 places can be filled in 5 ways (with the digits 0,2,4,6,8). $\therefore$ Number of possible licence plates of the given type is $(5 \times 5) \times (5 \times 5 \times 5 \times 5) = 5^6 = 15{,}625$.

**Problem 2.** QP Cars of a particular company come in 4 models, 12 colours, 3 engine sizes and 2 transmission types. (a) How many distinct cars of this company can be manufactured? (b) Of these, how many have the same colour?

**Solution.** (a) The number of distinct cars that can be manufactured is $4 \times 12 \times 3 \times 2 = 288$.
(b) For any chosen colour, the number of distinct cars that can be manufactured is $4 \times 3 \times 2 = 24$.

**Problem 3.** Find the number of 3-digit even numbers with no repeated digits.

**Solution.** Consider the number in the form $xyz$, where each of $x, y, z$ represents a digit. Since $xyz$ has to be even, $z$ has to be 0, 2, 4, 6 or 8.

- If $z = 0$ (1 choice only), then $x$ has 9 choices and $y$ has 8 choices.
- If $z = 2, 4, 6$ or 8 (4 choices), then $x$ has 8 choices and $y$ has 8 choices. [Note that $y$ can take 0, but $x$ cannot take 0.]

$\therefore$ The desired number is $(9 \times 8 \times 1) + (8 \times 8 \times 4) = 72 + 256 = 328$.

**Problem 4.** QP A bit is either 0 or 1. A byte is a sequence of 8 bits. Find (i) the number of bytes; (ii) the number of bytes that begin with 11 and end with 11; (iii) the number of bytes that begin with 11 and do not end with 11; (iv) the number of bytes that begin with 11 or end with 11.

**Solution.** (i) Since each byte contains 8 bits and each bit is a 0 or 1 (two choices), the number of bytes is $2 \times 2 \times \cdots \times 2$ (8 times) $= 2^8 = 256$.

(ii) In a byte beginning and ending with 11 (`11____11`), there are 4 empty positions. These can be filled in $2^4 = 16$ ways. $\therefore$ There are 16 bytes which begin and end with 11.

(iii) In a byte beginning with 11, there are 6 empty positions. These can be filled in $2^6 = 64$ ways, so there are 64 bytes that begin with 11. Since 16 bytes begin and end with 11, the number of bytes that begin with 11 but do not end with 11 is $64 - 16 = 48$.

(iv) The number of bytes that end with 11 is also 64 (as in (iii)), and the number of bytes that begin and end with 11 is 16. ∴ The number of bytes that begin with 11 or end with 11 is

$$64 + 64 - 16 = 112 \qquad [\because \text{ for any two finite sets } A, B: \ |A \cup B| = |A| + |B| - |A \cap B|].$$

Let $A$: bytes that begin with 11, $B$: bytes that end with 11. Then $A \cap B$: begin and end with 11; $A \cup B$: begin or end with 11. Part (iii) asks for $|A - B| = |A| - |A \cap B|$; part (iv) asks for $|A \cup B| = |A| + |B| - |A \cap B|$. (This is the simplest case of the Principle of Inclusion–Exclusion, Section 9.)

**Problem 5.** Find the number of proper divisors of 441000. (Proper divisors are all divisors excluding 1 and the number itself.)

**Solution.**

We have $441000 = 2^3 \cdot 3^2 \cdot 5^3 \cdot 7^2$ (Fundamental Theorem of Arithmetic).
∴ Every divisor of $n = 441000$ must be of the form $d = 2^p\, 3^q\, 5^r\, 7^s$, where $0 \le p \le 3,\ 0 \le q \le 2,\ 0 \le r \le 3,\ 0 \le s \le 2$.
So $p$ can be chosen in 4 ways, $q$ in 3 ways, $r$ in 4 ways and $s$ in 3 ways. Thus the total number of divisors $= 4 \times 3 \times 4 \times 3 = 144$.
The total number of proper divisors $= 144 - 2 = 142$ [$-2$ because a number is always divisible by 1 and by itself; so we leave out these 2 choices].

| | |
|---|---|
| 2 | 441000 |
| 2 | 220500 |
| 2 | 110250 |
| 3 | 55125 |
| 3 | 18375 |
| 5 | 6125 |
| 5 | 1225 |
| 5 | 245 |
| 7 | 49 |
| 7 | 7 |
| | 1 |

**Problem 6.** Licence plates consist of 2 letters followed by 4 digits. In how many ways can this be done if (i) no letter or digit can be repeated; (ii) repetitions are allowed; (iii) only vowels and even digits can be used (with repetitions)?

**Solution.** (i) There are 26 letters in English: the 1st place has 26 choices and the 2nd place has 25 choices (since no repetition is allowed). Similarly, the digits are 10 in number $(0, 1, \ldots, 9)$: the 1st place can be filled in 10 ways, the 2nd in 9, the 3rd in 8 and the 4th in 7 ways. ∴ Total number of ways $= 26 \times 25 \times 10 \times 9 \times 8 \times 7 = 32{,}76{,}000$.
(ii) Since repetitions are allowed: $(26 \times 26) \times (10 \times 10 \times 10 \times 10) = 26^2 \times 10^4 = 67{,}60{,}000$.
(iii) There are 5 vowels to be used, and the even digits are 0,2,4,6,8 (5 choices). ∴ $(5{\times}5){\times}(5{\times}5{\times}5{\times}5) = 5^6 = 15{,}625$.

**Problem 7.** Find the total number of positive integers that can be formed from the digits 1, 2, 3, 4 if no digit is repeated in any one integer.

**Solution.** Let $s_1, s_2, s_3, s_4$ denote the number of integers containing 1 digit, 2 digits, 3 digits and 4 digits respectively.

- There are 4 integers containing exactly 1 digit (1, 2, 3 or 4): $s_1 = 4$.
- A 2-digit integer without repetition can be chosen in $4 \times 3 = 12$ ways: $s_2 = 12$.
- A 3-digit integer without repetition can be chosen in $4 \times 3 \times 2 = 24$ ways: $s_3 = 24$.
- Similarly, $s_4 = 4 \times 3 \times 2 \times 1 = 24$.

Thus the required number is $s_1 + s_2 + s_3 + s_4 = 4 + 12 + 24 + 24 = 64$.

**Problem 8.** QP Determine the number of 6-digit integers (no leading 0's) in which (i) no digit is repeated; (ii) no digit is repeated and it is even; (iii) no digit is repeated and it is divisible by 5.

**Solution.** (i) The first place cannot be zero (9 choices); the second place can be zero (9 choices); and so on: $9 \times 9 \times 8 \times 7 \times 6 \times 5 = 1{,}36{,}080$.

(ii) There are 2 possibilities for the unit (last) place.
*Case 1:* If the last place is 0 (1 choice): $9 \times 8 \times 7 \times 6 \times 5 \times 1 = 15{,}120 = s_1$ (say).
*Case 2:* If the last place is not 0, i.e., the unit place takes 2,4,6,8 (4 choices), then the first place has 8 choices (no zero, and not the unit digit): $8 \times 8 \times 7 \times 6 \times 5 \times 4 = 53{,}760 = s_2$ (say).
Thus the required number $= s_1 + s_2 = 15{,}120 + 53{,}760 = 68{,}880$.

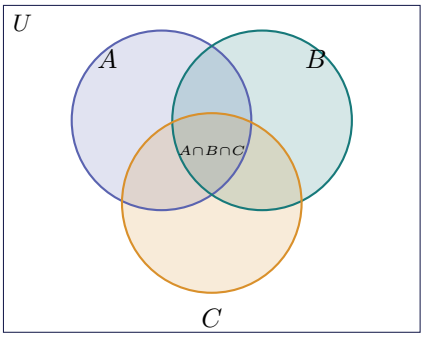
(iii) *Case 1:* If the unit place is 0: $9 \times 8 \times 7 \times 6 \times 5 \times 1 = 15{,}120 = s_3$.
*Case 2:* If the unit place is not zero, i.e., it is 5 (1 choice); the first place cannot be 0 or 5 (8 choices): $8{\times}8{\times}7{\times}6{\times}5{\times}1 = 13{,}440 = s_4$.
∴ Required number $= s_3 + s_4 = 15{,}120 + 13{,}440 = 28{,}560$.

**Practice Questions — Sum and Product Rules**

1. How many bit strings of length 10 are there? How many begin with 1 and end with 0?
2. A password is 6 to 8 characters long; each character is an uppercase letter or a digit, and each password must contain at least one digit. How many passwords are possible?
3. How many 4-digit numbers with distinct digits are divisible by 5?
4. How many positive divisors does 10800 have? QP
5. A restaurant offers 4 starters, 6 main courses and 3 desserts. How many 3-course meals are possible? How many meals if a customer may skip the starter?
6. How many functions are there from a set with $m$ elements to a set with $n$ elements? How many of them are one-to-one ($m \le n$)?

# 9 The Principle of Inclusion–Exclusion NEW

When we add $|A|+|B|$, the elements of $A \cap B$ are counted twice. Subtracting $|A \cap B|$ once corrects this. Generalising this "include, then exclude the overlap" idea gives the Principle of Inclusion–Exclusion (PIE).




Each element of $A \cup B \cup C$ is counted exactly **once** by

$$|A|+|B|+|C|-|A\cap B|-|A\cap C|$$
$$-|B\cap C|+|A\cap B\cap C|.$$

E.g. an element in all three sets is counted $3-3+1=1$ time.

**Principle of Inclusion–Exclusion** QP

For finite sets:

$$|A\cup B| = |A|+|B|-|A\cap B|$$
$$|A\cup B\cup C| = |A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$$

In general, for finite sets $A_1, A_2, \dots, A_n$:

$$\Big|\bigcup_{i=1}^{n} A_i\Big| = \sum_{i}|A_i| - \sum_{i<j}|A_i\cap A_j| + \sum_{i<j<k}|A_i\cap A_j\cap A_k| - \cdots + (-1)^{n+1}|A_1\cap A_2\cap\cdots\cap A_n|.$$

**Proof.** (General case.) Let $x$ belong to exactly $r \ge 1$ of the sets. On the right-hand side $x$ is counted $\binom{r}{1}$ times in the first sum, $\binom{r}{2}$ times in the second, and so on. So its total count is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1}\binom{r}{r} = 1 - \left[\binom{r}{0} - \binom{r}{1} + \cdots + (-1)^r\binom{r}{r}\right] = 1 - 0 = 1,$$

using the identity $\sum_{k}(-1)^k\binom{r}{k} = 0$ (Section 13, Problem 1(ii)). Hence every element of the union is counted exactly once. ∎

**Complement form ("none of the properties")**

If $N = |U|$ and $A_i$ is the set of elements having property $P_i$, the number of elements having **none** of the properties is

$$N(\overline{P_1}\,\overline{P_2}\cdots\overline{P_n}) = |\overline{A_1}\cap\overline{A_2}\cap\cdots\cap\overline{A_n}| = N - \sum|A_i| + \sum|A_i\cap A_j| - \cdots + (-1)^n|A_1\cap\cdots\cap A_n|.$$

**Useful fact:** the number of integers in $\{1, 2, \ldots, N\}$ divisible by $d$ is $\lfloor N/d \rfloor$. Integers divisible by both $a$ and $b$ are exactly those divisible by $\mathrm{lcm}(a, b)$.

## 9.1 Examples

**Example 1.** In a class of 50 students, 20 play cricket, 15 play football and 8 play both. How many play neither game?

**Solution.** $|C \cup F| = 20 + 15 - 8 = 27$. Neither $= 50 - 27 = \mathbf{23}$.

**Example 2.** QP How many integers from 1 to 1000 are divisible by 3, 5 or 7?

**Solution.** Let $A_3, A_5, A_7$ be the sets of integers in $\{1, \ldots, 1000\}$ divisible by 3, 5, 7.

$$|A_3| = \left\lfloor \tfrac{1000}{3} \right\rfloor = 333, \quad |A_5| = \left\lfloor \tfrac{1000}{5} \right\rfloor = 200, \quad |A_7| = \left\lfloor \tfrac{1000}{7} \right\rfloor = 142,$$

$|A_3 \cap A_5| = \left\lfloor \tfrac{1000}{15} \right\rfloor = 66$, $|A_3 \cap A_7| = \left\lfloor \tfrac{1000}{21} \right\rfloor = 47$, $|A_5 \cap A_7| = \left\lfloor \tfrac{1000}{35} \right\rfloor = 28$, $|A_3 \cap A_5 \cap A_7| = \left\lfloor \tfrac{1000}{105} \right\rfloor = 9$.

$|A_3 \cup A_5 \cup A_7| = (333 + 200 + 142) - (66 + 47 + 28) + 9 = 675 - 141 + 9 = \mathbf{543}$.

**Example 3.** Find the number of integers between 1 and 300 that are *not* divisible by any of 2, 3, 5.

**Solution.** $N = 300$. $|A_2| = 150$, $|A_3| = 100$, $|A_5| = 60$; $|A_2 \cap A_3| = 50$, $|A_2 \cap A_5| = 30$, $|A_3 \cap A_5| = 20$; $|A_2 \cap A_3 \cap A_5| = 10$.

$$N(\overline{A_2}\,\overline{A_3}\,\overline{A_5}) = 300 - (150 + 100 + 60) + (50 + 30 + 20) - 10 = 300 - 310 + 100 - 10 = \mathbf{80}.$$

**Example 4.** A survey of 2092 students found that 1232 studied Spanish, 879 French and 114 Russian; 103 studied both Spanish and French, 23 Spanish and Russian, and 14 French and Russian. If every student studied at least one of these languages, how many studied all three?

**Solution.** $|S \cup F \cup R| = 2092$. By PIE, $2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$, so $|S \cap F \cap R| = 2092 - 2085 = \mathbf{7}$.

**Example 5.** (*Euler's $\phi$ function.*) How many integers in $\{1, 2, \ldots, 100\}$ are relatively prime to 100?

**Solution.** $100 = 2^2 \cdot 5^2$, so an integer is co-prime to 100 iff it is divisible by neither 2 nor 5. $100 - (50 + 20) + 10 = \mathbf{40}$. In general, if $n = p_1^{e_1} \cdots p_k^{e_k}$, PIE gives

$$\phi(n) = n\Big(1 - \frac{1}{p_1}\Big)\Big(1 - \frac{1}{p_2}\Big)\cdots\Big(1 - \frac{1}{p_k}\Big); \qquad \phi(100) = 100 \cdot \tfrac{1}{2} \cdot \tfrac{4}{5} = 40.$$

**Example 6.** QP Find the number of integer solutions of $x_1 + x_2 + x_3 = 11$ with $0 \le x_1 \le 3$, $0 \le x_2 \le 4$, $0 \le x_3 \le 6$.

**Solution.** Without upper bounds, the number of non-negative solutions is $C(3 + 11 - 1, 11) = C(13, 2) = 78$ (Section 14). Let $P_1 : x_1 \ge 4$, $P_2 : x_2 \ge 5$, $P_3 : x_3 \ge 7$.

- $N(P_1)$: put $x_1 = y_1 + 4$, giving $y_1 + x_2 + x_3 = 7$: $C(9, 2) = 36$.
- $N(P_2)$: $x_1 + y_2 + x_3 = 6$: $C(8, 2) = 28$. $N(P_3)$: $x_1 + x_2 + y_3 = 4$: $C(6, 2) = 15$.
- $N(P_1P_2)$: sum $= 2$: $C(4, 2) = 6$. $N(P_1P_3)$: sum $= 0$: $C(2, 2) = 1$. $N(P_2P_3)$: sum $= -1$: 0. $N(P_1P_2P_3) = 0$.

Required number $= 78 - (36 + 28 + 15) + (6 + 1 + 0) - 0 = \mathbf{6}$.

**Example 7.** QP (*Onto functions.*) How many onto functions are there from a set with 6 elements to a set with 3 elements?

**Solution.** Total functions $= 3^6$. Let $A_i$ = functions that miss the $i$-th element of the codomain. $|A_i| = 2^6$, $|A_i \cap A_j| = 1^6$, $|A_1 \cap A_2 \cap A_3| = 0$.

$$\text{Onto} = 3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 - 0 = 729 - 192 + 3 = \mathbf{540}.$$

In general, the number of onto functions from an $m$-set to an $n$-set is $\displaystyle\sum_{k=0}^{n}(-1)^k\binom{n}{k}(n-k)^m$.

**Example 8.** QP (*Derangements.*) A *derangement* is a permutation of $1, 2, \ldots, n$ in which no element is in its original position. Find the number of derangements $D_n$, and evaluate $D_4$.

**Solution.** Let $A_i$ = permutations that fix $i$. Then $|A_{i_1} \cap \cdots \cap A_{i_k}| = (n-k)!$, and there are $\binom{n}{k}$ such intersections. By the complement form of PIE,

$$D_n = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \cdots + (-1)^n \binom{n}{n} 0! = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right].$$

$D_4 = 24\left[1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right] = 12 - 4 + 1 = \mathbf{9}$.

*Interpretation (hat-check problem):* if 4 students' hats are returned at random, the probability that nobody gets their own hat is $\frac{9}{24} = 0.375$. As $n \to \infty$, $D_n/n! \to e^{-1} \approx 0.368$.

**Practice Questions — Inclusion–Exclusion**

1. How many integers from 1 to 1000 are divisible by neither 5 nor 7?
2. Find the number of integers between 1 and 250 that are divisible by any of 2, 3, 5, 7. How many are divisible by none of them? QP
3. In a group of 60 students, 25 know C, 26 know Java and 26 know Python; 11 know C and Java, 9 know C and Python, 8 know Java and Python, and 3 know all three. How many know at least one language? How many know none?
4. How many onto functions are there from a 5-element set to a 3-element set? From a 7-element set to a 4-element set?
5. Find $D_5$ and $D_6$.
6. Find the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 20$ with $0 \le x_i \le 7$ for each $i$.
7. Find $\phi(360)$ using inclusion–exclusion.
8. How many permutations of the digits $0, 1, 2, \ldots, 9$ do not begin with 0 *and* do not end with 9? (Hint: complement form with two properties.)

# 10 The Pigeon-hole Principle NEW

**The Pigeon-hole Principle** QP

If $n+1$ (or more) pigeons are placed in $n$ pigeon-holes, then at least one pigeon-hole contains **two or more** pigeons.

**Proof.** If each of the $n$ holes contained at most one pigeon, there would be at most $n$ pigeons — a contradiction. ■

**The Generalised Pigeon-hole Principle** QP

If $N$ objects are placed into $k$ boxes, then at least one box contains at least $\lceil N/k \rceil$ objects.
Equivalently: if $m$ pigeons occupy $n$ holes, some hole has at least $\left\lfloor \frac{m-1}{n} \right\rfloor + 1$ pigeons.

**Proof.** Suppose every box has at most $\lceil N/k \rceil - 1$ objects. Since $\lceil N/k \rceil < \frac{N}{k} + 1$, the total is at most $k\big(\lceil N/k \rceil - 1\big) < k \cdot \frac{N}{k} = N$ — a contradiction. ■

**How to apply the principle:** (1) decide what the *pigeons* are (the objects chosen); (2) design the *holes* (categories) so that "two pigeons in one hole" gives the property you want; (3) check that pigeons $>$ holes. A useful corollary: the minimum number of objects that guarantees at least $r$ objects in some one of $k$ boxes is $k(r-1)+1$.

## 10.1 Examples

**Example 1.** Show that in any group of 13 people, at least two were born in the same month. In any group of 367 people, at least two share a birthday.

**Solution.** Pigeons = people, holes = 12 months; $13 > 12$. Similarly, there are at most 366 possible birthdays and $367 > 366$.

**Example 2.** A drawer contains 10 black and 10 white socks (unpaired). In the dark, how many socks must be drawn to be sure of (i) a matching pair; (ii) two black socks?

**Solution.** (i) Holes = 2 colours. Drawing 3 socks guarantees two of the same colour: **3**.
(ii) In the worst case all 10 white socks come first; then 2 more give two black: $10 + 2 = \mathbf{12}$.

**Example 3.** QP What is the minimum number of students in a class to be sure that at least 6 receive the same grade, if there are 5 possible grades (A, B, C, D, E)?

**Solution.** We need the least $N$ with $\lceil N/5 \rceil \geq 6$, i.e., $N = 5(6-1)+1 = \mathbf{26}$. With 25 students, 5 per grade is possible, so 26 is the minimum.

**Example 4.** QP Prove that among any $n+1$ integers, there are two whose difference is divisible by $n$.

**Solution.** Holes = the $n$ possible remainders $0, 1, \ldots, n-1$ on division by $n$ (Division Algorithm). With $n+1$ integers, two of them, say $a$ and $b$, have the same remainder. Then $a \equiv b \pmod{n}$, i.e., $n \mid (a-b)$. ■

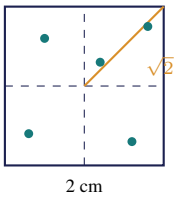
**Example 5.** QP Prove that if any 6 numbers are chosen from $\{1, 2, \ldots, 10\}$, then two of them add up to 11.

**Solution.** Form 5 holes: $\{1,10\}, \{2,9\}, \{3,8\}, \{4,7\}, \{5,6\}$. Each pair sums to 11 and every number lies in exactly one pair. Choosing 6 numbers (pigeons) from 5 holes forces two chosen numbers into the same pair. Their sum is 11. ■

**Example 6.** Prove that if 5 points are chosen inside a square of side 2 cm, then at least two of them are at a distance of at most $\sqrt{2}$ cm.

**Solution.**

Divide the square into 4 smaller squares of side 1 cm (holes). With 5 points (pigeons), some small square contains at least two of the points. The greatest distance between two points of a $1 \times 1$ square is its diagonal, $\sqrt{1^2+1^2} = \sqrt{2}$ cm. Hence those two points are at most $\sqrt{2}$ cm apart. ■




**Example 7.** QP Let $a_1, a_2, \ldots, a_n$ be any $n$ integers. Prove that there exist consecutive terms $a_{i+1}, \ldots, a_j$ whose sum is divisible by $n$.

**Solution.** Consider the $n$ partial sums $s_1 = a_1,\ s_2 = a_1 + a_2,\ \ldots,\ s_n = a_1 + \cdots + a_n$.
*Case 1:* some $s_j \equiv 0 \pmod{n}$. Then $a_1 + \cdots + a_j$ is divisible by $n$.
*Case 2:* no $s_j$ is $\equiv 0$. Then the $n$ sums have remainders in $\{1, \ldots, n-1\}$ — only $n-1$ holes. So two sums $s_i, s_j$ ($i < j$) have the same remainder, and $n \mid (s_j - s_i) = a_{i+1} + \cdots + a_j$. ■

**Example 8.** (*A link to Graph Theory.*) Prove that in any group of 6 people, there are either 3 mutual friends or 3 mutual strangers.

**Solution.** Fix a person $P$. The other 5 people fall into 2 holes: "friends of $P$" and "strangers to $P$". By the generalised principle, one hole has at least $\lceil 5/2 \rceil = 3$ people, say $X, Y, Z$ are friends of $P$ (the other case is symmetric). If any two of $X, Y, Z$ are friends, they form 3 mutual friends with $P$. Otherwise $X, Y, Z$ are 3 mutual strangers. ■
(In graph language: every red/blue colouring of the edges of $K_6$ contains a single-coloured triangle; the Ramsey number $R(3,3) = 6$.)

**Example 9.** Show that in any group of $n \geq 2$ people, there are two people with the same number of friends within the group.

**Solution.** Each person has $0, 1, \ldots, n-1$ friends — $n$ values. But the values 0 and $n-1$ cannot both occur (if someone is friends with everyone, nobody has 0 friends). So at most $n-1$ values are actually used by $n$ people, and two people share a value. ■

**Practice Questions — Pigeon-hole Principle**

1. Show that among 100 people, at least 9 were born in the same month.
2. How many cards must be drawn from a standard 52-card deck to guarantee (i) 3 cards of the same suit; (ii) 3 hearts?
3. India has 28 states and 8 Union Territories. What is the minimum number of students in an all-India class to guarantee that at least 4 come from the same state/UT?
4. Show that among any 7 integers, there are two whose difference is divisible by 6.
5. If 5 integers are chosen from $\{1, 2, \ldots, 8\}$, show that two of them add up to 9.
6. Show that if 5 points are placed in an equilateral triangle of side 1, two of them are within distance $\frac{1}{2}$.
7. Show that if any $n+1$ integers are chosen from $\{1, 2, \ldots, 2n\}$, two of them are consecutive. QP
8. A computer lab has 15 workstations and 10 servers. Each workstation is connected by a cable to some server; only one workstation may use a server at a time. What is the minimum number of cables needed to guarantee that any 10 workstations can simultaneously use different servers? (Challenge.)
9. (Challenge) Prove that every sequence of $n^2+1$ distinct real numbers contains a strictly increasing or strictly decreasing subsequence of length $n+1$.

# 11 Permutations

A linear arrangement of distinct objects is called a **permutation**. The number of arrangements of $r$ distinct objects out of $n$ objects is denoted by $P(n, r)$ or ${}^nP_r$, and

$$P(n,r) = n(n-1)(n-2)\cdots\big(n-(r-1)\big) = \frac{n!}{(n-r)!}.$$

**Note:**

1. The number of different arrangements of $n$ objects taken all $n$ at a time is $n!$. This is the number of permutations of $n$ distinct objects.
2. Suppose we wish to find the number of permutations that can be formed from a collection of $n$ objects of which $n_1$ are of one type, $n_2$ are of a 2nd type, $n_3$ of a 3rd type, …, $n_k$ are of the $k$-th type, with $n_1 + n_2 + \cdots + n_k = n$. Then the number of permutations of the $n$ objects taken all at a time is
$$\frac{n!}{n_1!\, n_2! \cdots n_k!}.$$
3. The number of circular arrangements of $n$ different objects is $(n-1)!$.

## 11.1 Problems

**Problem 1.** 5 men and 4 women sit in a row such that the women occupy the even places. How many such arrangements are possible?

**Solution.** The 5 men can be seated (in the odd places) in 5! ways and the 4 women can be placed in the even places in 4! ways. $\therefore$ Total number of arrangements $= 5! \times 4! = 120 \times 24 = 2880$.

**Problem 2.** Find $n$ if (i) $P(n, 2) = 90$; (ii) $P(n, 3) = 3P(n, 2)$.

**Solution.** (i) $P(n,2) = n(n-1) = 90 \Rightarrow n^2 - n - 90 = 0 \Rightarrow n^2 - 10n + 9n - 90 = 0 \Rightarrow n(n-10) + 9(n-10) = 0 \Rightarrow (n-10)(n+9) = 0 \Rightarrow n = 10$ or $-9$. $\therefore n = 10$ ($n$ must be positive).

(ii) $P(n,3) = 3P(n,2) \Rightarrow \dfrac{n!}{(n-3)!} = 3 \cdot \dfrac{n!}{(n-2)!} \Rightarrow \dfrac{(n-2)!}{(n-3)!} = 3 \Rightarrow \dfrac{(n-2)(n-3)!}{(n-3)!} = 3 \Rightarrow n-2 = 3 \Rightarrow n = 5$.

**Problem 3.** How many different strings (sequences) of length 4 can be formed using the letters of the word FLOWER?

**Solution.** FLOWER has 6 letters, all of which are distinct. The required number of strings is the number of permutations of these six letters chosen 4 at a time: $P(6,4) = \dfrac{6!}{(6-4)!} = \dfrac{6 \times 5 \times 4 \times 3 \times 2!}{2!} = 360.$

**Problem 4.** Find the number of distinguishable permutations of the letters in the following words: (i) BASIC (ii) CALCULUS (iii) MATHEMATICS (iv) STRUCTURES (v) ENGINEERING

**Solution.** (i) BASIC has 5 letters, all distinct. $\therefore$ Number of permutations $= 5! = 120$.
(ii) CALCULUS has 8 letters, of which 2 are C, 2 are L, 2 are U, and 1 each are A and S. $\therefore$ Required number $= \dfrac{8!}{2!\,2!\,2!\,1!\,1!} = 5040$.
(iii) MATHEMATICS has 11 letters: M = 2, A = 2, T = 2, H = 1, E = 1, I = 1, C = 1, S = 1. $\therefore$ Number $= \dfrac{11!}{2! \times 2! \times 2! \times (1!)^5} = 49{,}89{,}600$.
(iv) STRUCTURES has 10 letters: S = 2, T = 2, R = 2, U = 2, C = 1, E = 1. $\therefore$ Number $= \dfrac{10!}{2!\,2!\,2!\,2!\,1!\,1!} = 2{,}26{,}800$.
(v) ENGINEERING has 11 letters: E = 3, N = 3, G = 2, I = 2, R = 1. $\therefore$ Number $= \dfrac{11!}{3!\,3!\,2!\,2!\,1!} = 2{,}77{,}200$.

**Problem 5.** Find the number of permutations of the letters of the word MASSASAUGA. In how many of these are all four A's together? How many of them begin with S?

**Solution.** (i) The given word has 10 letters, of which 4 are A, 3 are S and 1 each are M, U, G. $\therefore$ Required number $= \dfrac{10!}{4!\,3!\,1!\,1!\,1!} = 25{,}200$.
(ii) If all A's are to be together, we treat all the A's as one single letter, say X = (AAAA). Then the letters to be permuted read X, S, S, S, M, U, G, which are 7 in number. $\therefore$ Number $= \dfrac{7!}{1!\,3!\,1!\,1!\,1!} = 840$.
(iii) For permutations beginning with S, there are nine positions to fill, with two S, four A and one each of M, U, G. $\therefore$ Number $= \dfrac{9!}{2!\,4!\,(1!)^3} = 7560$.

**Problem 6.** Find the number of permutations of the letters of the word MISSISSIPPI. How many of these begin with an I? How many of these begin and end with an S?

**Solution.** (i) The given word has 11 letters: 4 I's, 4 S's, 2 P's and 1 M. $\therefore$ Number $= \dfrac{11!}{4!\,4!\,2!\,1!} = 34{,}650$.
(ii) For permutations beginning with I, there are 10 positions to fill, with 3 I, 4 S, 2 P and 1 M. $\therefore$ Number $= \dfrac{10!}{3!\,4!\,2!\,1!} = 12{,}600$.
(iii) For permutations beginning and ending with S, there are 9 positions to fill, with 2 S, 4 I, 2 P and 1 M. $\therefore$ Number $= \dfrac{9!}{2!\,4!\,2!\,1!} = 3780$.

**Problem 7.** (a) How many arrangements are there for all letters in the word SOCIOLOGICAL? (b) In how many of these arrangements (i) are A and G adjacent? (ii) are all the vowels adjacent?

**Solution.** (a) The word has 12 letters: 3 O; 2 each of C, I, L; and 1 each of S, A, G. $\therefore$ Number $= \dfrac{12!}{3!\,2!\,2!\,2!\,(1!)^3} = 99{,}79{,}200$.

(b)(i) If A and G are to be adjacent, treat A and G together as a single letter X, so that there are 3 O's, 2 C's, 2 I's, 2 L's, 1 S and 1 X, totalling 11 letters. These can be arranged in $\dfrac{11!}{3!\,2!\,2!\,2!\,1!\,1!} = 8{,}31{,}600$ ways. Also, A and G can be arranged among themselves in $2! = 2$ ways. $\therefore$ Total $= 8{,}31{,}600 \times 2 = 16{,}63{,}200$.

(ii) If all the vowels are to be adjacent, treat all the vowels (A, O, I) as a single letter Y, so that there are 2 C's, 2 L's, 1 S, 1 G and 1 Y, totalling 7 letters. These can be arranged in $\dfrac{7!}{2!\,2!\,1!\,1!\,1!} = 1260$ ways. Since the word contains 3 O's, 2 I's and 1 A (totalling 6), the letters within Y can be arranged in $\dfrac{6!}{3!\,2!\,1!} = 60$ ways. $\therefore$ Total $= 1260 \times 60 = 75{,}600$.

**Problem 8.** How many six-digit numbers can one make using the digits 1, 3, 3, 7, 7, 8?

**Solution.** Here $n = 6$, with one 1, two 3's, two 7's and one 8. Number $= \dfrac{6!}{1!\,2!\,2!\,1!} = \dfrac{720}{4} = 180$.

**Problem 9.** How many positive integers $n$ can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want $n$ to exceed 50,00,000?

**Solution.** Let $n = x_1x_2x_3x_4x_5x_6x_7$. Since $n$ exceeds 50,00,000, $x_1$ can be 5, 6 or 7.

- If $x_1 = 5$, the remaining 6 digits contain two 4's and one each of 3, 5, 6, 7: $\dfrac{6!}{2!} = 360$.
- If $x_1 = 6$, the remaining 6 digits contain two 4's, two 5's and one each of 3, 7: $\dfrac{6!}{2!\,2!} = 180$.
- If $x_1 = 7$, the remaining 6 digits contain two 4's, two 5's and one each of 3, 6: $\dfrac{6!}{2!\,2!} = 180$.

$\therefore$ By the sum rule, the number of such $n$ is $360 + 180 + 180 = 720$.

**Problem 10.** In how many ways can 6 men and 6 women be seated in a row (i) if any person may sit next to any other? (ii) if men and women must occupy alternate seats?

**Solution.** (i) No distinction is made: $12! = 47{,}90{,}01{,}600$.
(ii) Men in the odd places and women in the even places: $6! \times 6! = 5{,}18{,}400$. Similarly, men in even places and women in odd places: $5{,}18{,}400$. $\therefore$ Total $= 10{,}36{,}800$.

**Problem 11.** For non-negative integers $n$ and $r$, if $n + 1 > r$, prove that $P(n+1, r) = \dfrac{n+1}{n+1-r} P(n, r)$.

**Solution.** We have $P(n, r) = \dfrac{n!}{(n-r)!}$ and $P(n+1, r) = \dfrac{(n+1)!}{(n+1-r)!}$.

$$\therefore \frac{P(n+1,r)}{P(n,r)} = \frac{(n+1)!}{(n+1-r)!} \times \frac{(n-r)!}{n!} = \frac{(n+1)\cdot n!}{n!} \times \frac{(n-r)!}{(n+1-r)(n-r)!} = \frac{n+1}{n+1-r}.$$

Thus $P(n+1, r) = \left(\dfrac{n+1}{n+1-r}\right) P(n, r)$. ■

**Problem 12.** 4 different Mathematics books, 5 different CO books and 2 different GTC books are to be arranged in a shelf. How many different arrangements are possible if (a) the books in each particular subject must all be together? (b) only the Mathematics books must be together?

**Solution.** (a) Mathematics books: 4! ways; CO books: 5!; GTC books: 2!; the 3 groups: 3!. $\therefore$ Number $= 4! \times 5! \times 2! \times 3! = 34{,}560$.
(b) Treat the 4 Mathematics books as one book. Then there are 8 books, arranged in 8! ways, and the Mathematics books among themselves in 4! ways. $\therefore$ Number $= 8! \times 4! = 9{,}67{,}680$.

**Practice Questions — Permutations**

1. Find $n$ if $P(n, 4) = 42\, P(n, 2)$.
2. Find the number of arrangements of the letters of ASSASSINATION.
3. In how many ways can the letters of COMPUTER be arranged so that the vowels are together?
4. In how many ways can 7 people sit around a round table? In how many of these are two particular people *not* adjacent?
5. How many 5-digit numbers greater than 30,000 can be formed from the digits 1, 2, 3, 4, 5 without repetition?
6. How many permutations of the letters ABCDEFGH contain the string ABC?

# 12 Combinations

Selecting a set of $r$ objects from a set of $n$ objects ($n \geq r$) without regard to order is called a **combination** of $r$ objects. The total number of combinations of $r$ different objects that can be selected from $n$ different objects is denoted by $C(n, r)$ or ${}^nC_r$ and is given by

$${}^nC_r = \frac{n!}{(n-r)!\,r!}.$$

**Note:** ${}^nC_r = {}^nC_{n-r}$. Also $P(n,r) = r! \cdot C(n,r)$: an arrangement = a selection followed by ordering it.

## 12.1 Problems

**Problem 1.** How many committees of 5 members with one chairperson can be selected from 12 persons?

**Solution.** 1 chairperson out of 12 can be selected in 12 ways. The remaining 4 members out of 11 persons can be selected in ${}^{11}C_4$ ways. $\therefore$ Required number $= 12 \times {}^{11}C_4 = 12 \times 330 = 3960$.

**Problem 2.** A certain question paper contains 3 parts A, B, C with 4 questions in Part A, 5 in Part B and 6 in Part C. It is required to answer 7 questions, selecting at least 2 from each part. In how many different ways can a student select his questions?

**Solution.**

| Part | No. of questions | Case (1) | Case (2) | Case (3) |
|---|---|---|---|---|
| A | 4 | 2 | 2 | 3 |
| B | 5 | 2 | 3 | 2 |
| C | 6 | 3 | 2 | 2 |
| No. of ways | | ${}^4C_2 \times {}^5C_2 \times {}^6C_3$ $(S_1 = 1200)$ | ${}^4C_2 \times {}^5C_3 \times {}^6C_2$ $(S_2 = 900)$ | ${}^4C_3 \times {}^5C_2 \times {}^6C_2$ $(S_3 = 600)$ |

$\therefore$ Total number of ways $= S_1 + S_2 + S_3 = 1200 + 900 + 600 = 2700$.

**Problem 3.** Find the number of committees of 5 that can be selected from 7 men and 5 women, if the committee is to consist of at least 1 man and at least 1 woman.

**Solution.**

| Men | Women | No. of committees of 5 |
|---|---|---|
| 1 | 4 | ${}^7C_1 \times {}^5C_4 = 35$ |
| 2 | 3 | ${}^7C_2 \times {}^5C_3 = 210$ |
| 3 | 2 | ${}^7C_3 \times {}^5C_2 = 350$ |
| 4 | 1 | ${}^7C_4 \times {}^5C_1 = 175$ |

$\therefore$ Required number $= 35{+}210{+}350{+}175 = 770$. (*Check by complement:* ${}^{12}C_5 - {}^7C_5 - {}^5C_5 = 792 - 21 - 1 = 770$.)

**Problem 4.** A woman has 11 close relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following situations: (i) there is no restriction on the choice; (ii) 2 particular persons will not attend separately; (iii) 2 particular persons will not attend together?

**Solution.** (i) ${}^{11}C_5 = 462$ ways.
(ii) Let A and B be the 2 particular persons. (a) If both A and B are invited: ${}^9C_3 = 84$. (b) If neither is invited: ${}^9C_5 = 126$. $\therefore$ Total $= 84 + 126 = 210$.
(iii) (a) A invited, B not: ${}^9C_4 = 126$. (b) B invited, A not: ${}^9C_4 = 126$. (c) Neither invited: ${}^9C_5 = 126$. $\therefore$ Total $= 378$.

**Problem 5.** Find the number of arrangements of all the letters of the word TALLAHASSEE. How many of these arrangements have no adjacent A's?

**Solution.** (i) The word has 11 letters: 3 A's; 2 each of L, S, E; and 1 each of T and H. $\therefore$ Number $= \dfrac{11!}{3!\,(2!)^3\,(1!)^2} =$ 8,31,600.
(ii) If the A's are disregarded, the remaining 8 letters can be arranged in $\dfrac{8!}{(2!)^3(1!)^2} = 5040$ ways. In each of these arrangements, there are 9 possible locations for the 3 A's (the gaps marked ↑):

↑ T ↑ L ↑ L ↑ H ↑ S ↑ S ↑ E ↑ E ↑

These locations can be chosen in ${}^9C_3$ ways. $\therefore$ Number with no adjacent A's $= 5040 \times {}^9C_3 = 5040 \times 84 = 4{,}23{,}360$.

**Problem 6.** Find how many distinct 4-digit integers one can make from the digits 1, 3, 3, 7, 7, 8.

**Solution.** The following 3 cases arise:
(i) All distinct digits (1, 3, 7, 8): $4! = 24$ ways.
(ii) 2 pairs of identical digits (3, 3, 7, 7): $\dfrac{4!}{2!\,2!} = 6$ ways.
(iii) 1 pair of identical digits: the possible digit sets are $\{3,3,1,7\}$, $\{3,3,1,8\}$, $\{3,3,7,8\}$, $\{7,7,1,3\}$, $\{7,7,1,8\}$, $\{7,7,3,8\}$, which are 6 in number. This gives $\dfrac{4!}{2!\,1!\,1!} \times 6 = 72$ ways.
$\therefore$ Total number of ways $= 24 + 6 + 72 = 102$.

**Problem 7.** A man has 15 close friends, of whom 6 are women. In how many ways can he invite 3 or more of these friends to a party if he wants the same number of men (including himself) as women?

**Solution.** Out of the 15 friends, 6 are women and 9 are men.

| Women | Men (+ himself in all cases) | No. of ways |
|---|---|---|
| 2 | 1 | ${}^6C_2 \times {}^9C_1 = 135$ |
| 3 | 2 | ${}^6C_3 \times {}^9C_2 = 720$ |
| 4 | 3 | ${}^6C_4 \times {}^9C_3 = 1260$ |
| 5 | 4 | ${}^6C_5 \times {}^9C_4 = 756$ |
| 6 | 5 | ${}^6C_6 \times {}^9C_5 = 126$ |

$\therefore$ Total number of ways $= 135 + 720 + 1260 + 756 + 126 = 2997$.

**Problem 8.** How many bytes contain (i) exactly two 1's (ii) exactly four 1's (iii) exactly six 1's (iv) at least six 1's?

**Solution.** 1 byte = 8 bits and each bit is a 0 or 1.
(i) The positions of the two 1's can be chosen in ${}^8C_2$ ways and the remaining 6 positions are filled by zeros (1 choice).
$\therefore$ Number $= {}^8C_2 \times 1^6 = 28$.
(ii) ${}^8C_4 \times 1^4 = 70$. (iii) ${}^8C_6 \times 1^2 = 28$.
(iv) six 1's $\Rightarrow {}^8C_6$; seven 1's $\Rightarrow {}^8C_7$; eight 1's $\Rightarrow {}^8C_8$. $\therefore$ Number $= 28 + 8 + 1 = 37$.

**Problem 9.** Prove the following identities: (i) $C(n, r-1) + C(n, r) = C(n+1, r)$ (Pascal's identity); (ii) $C(m, 2) + C(n, 2) = C(m+n, 2) - mn$.

**Solution.** (i)

$$\begin{aligned} C(n,r-1) + C(n,r) &= \frac{n!}{(n-r+1)!\,(r-1)!} + \frac{n!}{(n-r)!\,r!} = \frac{n!}{(n-r+1)(n-r)!\,(r-1)!} + \frac{n!}{(n-r)!\,r\,(r-1)!} \\ &= \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{1}{n-r+1} + \frac{1}{r}\right] = \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{r+n-r+1}{r(n-r+1)}\right] \\ &= \frac{n!}{(n-r)!\,(r-1)!} \times \frac{(n+1)}{r(n-r+1)} = \frac{(n+1)!}{r!\,(n-r+1)!} = C(n+1, r). \end{aligned}$$

(ii)

$$\begin{aligned} C(m,2) + C(n,2) &= \frac{m!}{(m-2)!\,2!} + \frac{n!}{(n-2)!\,2!} = \frac{m(m-1)}{2} + \frac{n(n-1)}{2} = \frac{1}{2}\left[m^2 + n^2 - m - n\right] \\ &= \frac{1}{2}\left[(m+n)^2 - 2mn - m - n\right] = \frac{1}{2}\left[(m+n)^2 - (m+n)\right] - mn \\ &= \frac{1}{2}(m+n)(m+n-1) - mn = \frac{(m+n)!}{2!\,(m+n-2)!} - mn = C(m+n, 2) - mn. \end{aligned}$$

## Practice Questions — Combinations

1. A committee of 5 is to be formed from 8 men and 6 women. In how many ways can this be done if it must contain (i) exactly 3 women; (ii) at least 3 women?
2. There are 12 points in a plane, no three collinear. How many (i) straight lines; (ii) triangles can be formed?
3. Find the number of diagonals of a polygon with $n$ sides. For which $n$ does it equal $n$?
4. How many bytes contain exactly three 1's? At most two 1's?

5. A student must answer 10 out of 13 questions. How many choices does he have if he must answer at least 3 of the first 5 questions? QP
6. Prove that $C(n,r)\cdot C(r,k) = C(n,k)\cdot C(n-k,\, r-k)$ for $0 \le k \le r \le n$.

# 13 Binomial and Multinomial Theorems

A basic property of $C(n,r)$ is that it is the coefficient of $x^{n-r}y^r$ in the expansion of $(x+y)^n$, where $x$ and $y$ are any two real numbers:

$$(x+y)^n = x^n + {}^nC_1 x^{n-1}y + {}^nC_2 x^{n-2}y^2 + \cdots + {}^nC_{n-1}xy^{n-1} + y^n = \sum_{r=0}^{n} {}^nC_r\, x^{n-r}y^r. \tag{1}$$

Since ${}^nC_r = {}^nC_{n-r}$,

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^r y^{n-r}. \tag{2}$$

Thus ${}^nC_r$ is the coefficient of $x^r y^{n-r}$ in the expansion of $(x+y)^n$. ${}^nC_r$ is also denoted by $\binom{n}{r}$. Therefore

$$(1) \Rightarrow (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r}y^r, \qquad (2) \Rightarrow (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}. \tag{3}$$

**Binomial Theorem**

Expression (3) is called the Binomial Theorem for a positive integral index $n$. The numbers $\binom{n}{r}$ for $r = 0, 1, 2, \ldots, n$, namely $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}$, are called the **binomial coefficients**.

The generalisation of the Binomial Theorem is known as the Multinomial Theorem.

**Multinomial Theorem**

For positive integers $n$ and $k$, the coefficient of $x_1^{n_1}x_2^{n_2}\cdots x_k^{n_k}$ in the expansion of $(x_1 + x_2 + \cdots + x_k)^n$ is

$$\frac{n!}{n_1!\, n_2! \cdots n_k!},$$

where each $n_i$ is a non-negative integer $\le n$ and $n_1 + n_2 + \cdots + n_k = n$.

**Note:**

1. The general term in the expansion of $(x_1 + x_2 + \cdots + x_k)^n$ is $\dfrac{n!}{n_1!\, n_2! \cdots n_k!} x_1^{n_1}x_2^{n_2}\cdots x_k^{n_k}$ (alternate statement of the Multinomial Theorem).
2. The expression $\dfrac{n!}{n_1!\, n_2! \cdots n_k!}$ is also written as $\dbinom{n}{n_1, n_2, \ldots, n_k}$.
3. The Multinomial Theorem can also be stated as $(x_1+x_2+\cdots+x_k)^n = \displaystyle\sum_{n_i} \binom{n}{n_1, n_2, \ldots, n_k} x_1^{n_1}x_2^{n_2}\cdots x_k^{n_k}$.

## 13.1 Problems

**Problem 1.** Prove the following identities for a positive integer $n$: (i) $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$; (ii) $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n\binom{n}{n} = 0$.

**Solution.** The Binomial Theorem gives $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$.
(i) Taking $x = 1,\ y = 1$: $2^n = \sum_{r=0}^{n} \binom{n}{r}$, i.e., $2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$.
(ii) Taking $x = -1,\ y = 1$: $0 = \sum_{r=0}^{n} (-1)^r \binom{n}{r}$, i.e., $0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n\binom{n}{n}$.

**Problem 2.** Compute the following: (i) $\binom{12}{5,3,2,2}$ (ii) $\binom{8}{4,2,2,0}$ (iii) $\binom{10}{5,3,2,2}$

**Solution.** (i) $\binom{12}{5,3,2,2} = \dfrac{12!}{5!\,3!\,2!\,2!} = 1{,}66{,}320.$ (ii) $\binom{8}{4,2,2,0} = \dfrac{8!}{4!\,2!\,2!\,0!} = 420.$
(iii) Meaningless, since $5+3+2+2 = 12 \neq n = 10$.

**Problem 3.** Find the sum of all coefficients in the expansion of (a) $(x+y)^n$ (b) $(x_1 + x_2 + \cdots + x_k)^n$.

**Solution.** (a) Taking $x = 1,\ y = 1$ in $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$ gives $2^n = \sum_{r=0}^{n} \binom{n}{r}$. Thus the sum of all coefficients in $(x+y)^n$ is $2^n$.
(b) By the Multinomial Theorem, $\sum_{n_i} \binom{n}{n_1,\ldots,n_k} x_1^{n_1} \cdots x_k^{n_k} = (x_1 + \cdots + x_k)^n$. Taking $x_1 = x_2 = \cdots = x_k = 1$,

$$\sum_{n_i} \binom{n}{n_1, n_2, \ldots, n_k} = \underbrace{(1+1+\cdots+1)}_{k \text{ times}}{}^n = k^n.$$

Thus the sum of all coefficients in $(x_1 + x_2 + \cdots + x_k)^n$ is $k^n$.

**Problem 4.** Obtain the sum of all the coefficients in the expansion of (i) $(x+y)^{10}$ (ii) $(x+y+z)^{12}$ (iii) $(x+y+z+w)^5$ (iv) $(2s - 3t + 5u + 6v - 11w + 3x + 2y)^{10}$.

**Solution.** (i) $2^{10}$. (ii) $3^{12}$. (iii) $4^5$.
(iv) By the Multinomial Theorem,

$$\begin{aligned}(2s - 3t + 5u + 6v - 11w + 3x + 2y)^{10} &= \sum_{n_i} \binom{10}{n_1, \ldots, n_7} (2s)^{n_1} (-3t)^{n_2} (5u)^{n_3} (6v)^{n_4} (-11w)^{n_5} (3x)^{n_6} (2y)^{n_7} \\ &= \sum_{n_i} \binom{10}{n_1, \ldots, n_7} 2^{n_1} (-3)^{n_2} 5^{n_3} 6^{n_4} (-11)^{n_5} 3^{n_6} 2^{n_7} \\ &\quad \times s^{n_1} t^{n_2} u^{n_3} v^{n_4} w^{n_5} x^{n_6} y^{n_7}.\end{aligned}$$

Taking $s = t = u = v = w = x = y = 1$, the sum of the coefficients is $(2 - 3 + 5 + 6 - 11 + 3 + 2)^{10} = 4^{10}$.

**Problem 5.** Find the coefficient of (i) $x^3y^9$ in the expansion of $(x+2y)^{12}$; (ii) $x^0$ in the expansion of $\left(3x^2 - \frac{2}{x}\right)^{15}$; (iii) $x^{12}$ in the expansion of $x^3(1-2x)^{10}$; (iv) $x^k$ in the expansion of $(1+x+x^2)(1+x)^n$, where $n$ is a positive integer and $0 \leq k \leq n+2$.

**Solution.** (i) $(x+2y)^{12} = \sum_{r=0}^{12} \binom{12}{r} x^r (2y)^{12-r} = \sum_{r=0}^{12} \binom{12}{r} 2^{12-r} x^r y^{12-r}$. Taking $r = 3$, the coefficient of $x^3y^9$ is $\binom{12}{3} 2^9 = \dfrac{12!}{9!\,3!} \cdot 2^9 = 1{,}12{,}640.$
(ii)

$$\left(3x^2 - \frac{2}{x}\right)^{15} = \sum_{r=0}^{15} \binom{15}{r} (3x^2)^r \left(-\frac{2}{x}\right)^{15-r} = \sum_{r=0}^{15} \binom{15}{r} 3^r (-2)^{15-r} x^{2r} x^{r-15} = \sum_{r=0}^{15} \binom{15}{r} 3^r (-2)^{15-r} x^{3r-15}.$$

Taking $r = 5$ $(3r - 15 = 0)$, the coefficient of $x^0$ is $\binom{15}{5} 3^5 (-2)^{10} = {}^{15}C_5 \cdot 3^5 \cdot 2^{10} = 74{,}72{,}42{,}496.$

(iii) $(1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} (-2x)^r,\ \therefore x^3(1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} (-2)^r x^{r+3}$. Taking $r = 9$, the coefficient of $x^{12}$ is $\binom{10}{9}(-2)^9 = 10 \times (-512) = -5{,}120.$

(iv) $(1+x+x^2)(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r + \sum_{r=0}^{n} \binom{n}{r} x^{r+1} + \sum_{r=0}^{n} \binom{n}{r} x^{r+2}$. The coefficient of $x^k$ is obtained by taking $r = k$ in the 1st sum, $r = k-1$ in the 2nd and $r = k-2$ in the 3rd: it is $\binom{n}{k} + \binom{n}{k-1} + \binom{n}{k-2}$. For example, the coefficient of $x^{15}$ is $\binom{n}{15} + \binom{n}{14} + \binom{n}{13}$.

**Problem 6.** Determine the coefficient of (i) $xyz^5$ in $(x+y+z)^7$; (ii) $x^2y^2z^3$ in $(3x - 2y - 4z)^7$; (iii) $x^{11}y^4z^2$ in $(2x^3 - 3xy^2 + z^2)^6$; (iv) $xyz^{-2}$ in $(x - 2y + 3z^{-1})^4$; (v) $a^2b^3c^2d^5$ in $(a + 2b - 3c + 2d + 5)^{16}$.

**Solution.** (i) The general term in $(x+y+z)^7$ is $\binom{7}{n_1,n_2,n_3} x^{n_1} y^{n_2} z^{n_3}$. For $n_1 = 1, n_2 = 1, n_3 = 5$: $\binom{7}{1,1,5} = \dfrac{7!}{1!\,1!\,5!} = 42$. The coefficient of $xyz^5$ is **42**.

(ii) The general term in $(3x - 2y - 4z)^7$ is $\binom{7}{n_1,n_2,n_3} (3x)^{n_1} (-2y)^{n_2} (-4z)^{n_3}$. For $n_1 = 2, n_2 = 2, n_3 = 3$: the coefficient of $x^2y^2z^3$ is $\binom{7}{2,2,3} \times 9 \times 4 \times (-64) = -2304 \times \dfrac{7!}{2!\,2!\,3!} = -4{,}83{,}840.$

(iii) The general term in $(2x^3 - 3xy^2 + z^2)^6$ is

$$\binom{6}{n_1, n_2, n_3}(2x^3)^{n_1}(-3xy^2)^{n_2}(z^2)^{n_3} = \binom{6}{n_1, n_2, n_3}2^{n_1}(-3)^{n_2}x^{3n_1+n_2}y^{2n_2}z^{2n_3}.$$

For $x^{11}y^4z^2$: $2n_2 = 4 \Rightarrow n_2 = 2$; $3n_1 + 2 = 11 \Rightarrow n_1 = 3$; $2n_3 = 2 \Rightarrow n_3 = 1$ (and indeed $3 + 2 + 1 = 6$). The coefficient is $2^3(-3)^2\binom{6}{3,2,1} = 8 \times 9 \times \dfrac{6!}{3!\,2!\,1!} = 4{,}320$.

(iv) The general term in $(x - 2y + 3z^{-1})^4$ is $\binom{4}{n_1,n_2,n_3}(-2)^{n_2}3^{n_3}x^{n_1}y^{n_2}z^{-n_3}$. For $n_1 = 1, n_2 = 1, n_3 = 2$: the coefficient of $xyz^{-2}$ is $-2 \cdot 9 \times \binom{4}{1,1,2} = -18 \times \dfrac{4!}{1!\,1!\,2!} = -216$.

(v) The general term in $(a+2b-3c+2d+5)^{16}$ is $\binom{16}{n_1,\ldots,n_5}a^{n_1}(2b)^{n_2}(-3c)^{n_3}(2d)^{n_4}5^{n_5}$. For $n_1 = 2, n_2 = 3, n_3 = 2, n_4 = 5$: $n_5 = 16 - (2 + 3 + 2 + 5) = 4$. The coefficient of $a^2b^3c^2d^5$ is

$$2^3 \cdot (-3)^2 \cdot 2^5 \cdot 5^4 \times \binom{16}{2, 3, 2, 5, 4} = 14{,}40{,}000 \times \frac{16!}{2!\,3!\,2!\,5!\,4!} = 14{,}40{,}000 \times 30{,}27{,}02{,}400 = 4{,}35{,}89{,}14{,}56{,}00{,}000.$$

**Homework**

Find the coefficient of:

1. $x^5y^2$ in the expansion of $(2x - 3y)^7$.
2. $x^3z^4$ in the expansion of $(x + y + z)^7$.
3. $x^3y^3z^2$ in the expansion of $(2x - 3y + 5z)^8$.
4. $w^3x^2yz^2$ in the expansion of $(2w - x + 3y - 2z)^8$.
5. $x_1^2x_3x_4^3x_5^4$ in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$.
6. ( NEW ) Prove that $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n\,2^{n-1}$. (Hint: differentiate $(1 + x)^n$.)

## 14 Combinations with Repetitions

Suppose we have to select a combination of $r$ objects, with repetitions, from a set of $n$ distinct objects. Then the number of such selections is given by

$$C(n + r - 1,\, r) = \frac{(n + r - 1)!}{r!\,(n - 1)!} = C(r + n - 1,\, n - 1). \tag{1}$$

i.e., $C(n + r - 1, r)$ represents the number of combinations of $n$ distinct objects, taken $r$ at a time, with repetitions allowed.

**Why? (Stars and bars.)** A selection is described by $r$ stars ($\star$) and $n - 1$ bars ($|$) that separate the $n$ types. E.g. with $n = 3$ types and $r = 5$: $\star\star \mid \star \mid \star\star$ means 2 of type 1, 1 of type 2, 2 of type 3. Every arrangement of $r$ stars and $n - 1$ bars gives exactly one selection, so the count is $\binom{n-1+r}{r}$.

**Note:**

1. Eq. (1) represents the number of ways in which $r$ identical objects can be distributed among $n$ distinct containers (each container may receive $0, 1, 2, 3, \ldots$ objects).
2. Eq. (1) also represents the number of non-negative integer solutions of the equation $x_1 + x_2 + \cdots + x_n = r$, i.e., with $x_i \geq 0$ for $i = 1, 2, \ldots, n$. [A non-negative integer solution of $x_1 + x_2 + \cdots + x_n = r$ is an $n$-tuple $(x_1, x_2, \ldots, x_n)$ of non-negative integers whose sum is $r$.]
3. **How to identify $n$ and $r$:** In an ordinary combination ($n \geq r$) we select $r$ out of $n$. In a combination with repetition, $r$ may be greater than $n$: $n$ is the number of distinct places (containers / variables) and $r$ is the number of identical items to be distributed (the sum to be reached).
4. When some variables have *upper* bounds, combine this formula with Inclusion–Exclusion (Section 9, Example 6).

## 14.1 Problems

**Problem 1.** In how many ways can we distribute 10 identical marbles among 6 distinct containers?

**Solution.** Here $r = 10,\ n = 6$. $\therefore$ Required number $= C(6 + 10 - 1, 10) = C(15, 10) = \dfrac{15!}{10!\,5!} = 3003$.

**Problem 2.** Find the number of ways of placing 8 identical balls in 5 numbered boxes.

**Solution.** Here $r = 8,\ n = 5$. Required number $= C(5 + 8 - 1, 8) = C(12, 8) = \dfrac{12!}{8!\,4!} = 495$.

**Problem 3.** A bag contains coins of 7 different denominations, with at least 1 dozen coins in each denomination. In how many ways can we select a dozen coins from the bag?

**Solution.** Here $r = 12,\ n = 7$. Number of ways $= C(7 + 12 - 1, 12) = C(18, 12) = \dfrac{18!}{12!\,6!} = 18{,}564$.

**Problem 4.** Find the number of non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 8$.

**Solution.** Here $r = 8,\ n = 5$. Required number $= C(12, 8) = 495$.

**Problem 5.** Find the number of non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 7$.

**Solution.** Here $n = 4,\ r = 7$. Required number $= C(4 + 7 - 1, 7) = C(10, 7) = \dfrac{10!}{7!\,3!} = 120$.

**Problem 6.** Find the number of distinct terms in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{16}$.

**Solution.** The general term is $\binom{16}{n_1,\ldots,n_5} x_1^{n_1} x_2^{n_2} x_3^{n_3} x_4^{n_4} x_5^{n_5}$, where the $n_i$ are non-negative integers with sum 16. $\therefore$ Number of distinct terms = number of non-negative integer solutions of $n_1 + n_2 + n_3 + n_4 + n_5 = 16 = C(5 + 16 - 1, 16) = C(20, 16) = 4845$.

**Problem 7.** Find the number of non-negative integer solutions of the inequality $x_1 + x_2 + \cdots + x_6 < 10$.

**Solution.** Given $x_1 + \cdots + x_6 < 10$, i.e., $x_1 + \cdots + x_6 \le 9$. Write $x_1 + x_2 + \cdots + x_6 = 9 - x_7$, so that $x_7$ is a non-negative integer (a "slack" variable; e.g. $2 + 3 < 10 \Rightarrow 2 + 3 = 9 - 4$). $\therefore$ The required number is the number of non-negative solutions of $x_1 + x_2 + \cdots + x_7 = 9$, which is $C(7 + 9 - 1, 9) = C(15, 9) = 5005$.

**Problem 8.** Find the number of distinct terms in the expansion of $(w + x + y + z)^{10}$.

**Solution.** The number of distinct terms = number of non-negative integer solutions of $n_1 + n_2 + n_3 + n_4 = 10 = C(4 + 10 - 1, 10) = C(13, 10) = 286$.

**Problem 9.** Find the number of non-negative integer solutions of the inequality $x_1 + x_2 + \cdots + x_5 \le 19$.

**Solution.** $x_1 + x_2 + \cdots + x_5 \le 19 \Rightarrow x_1 + \cdots + x_5 = 19 - x_6$ with $x_6 \ge 0$. $\therefore$ Required number = number of non-negative integer solutions of $x_1 + \cdots + x_6 = 19 = C(6 + 19 - 1, 19) = C(24, 19) = 42{,}504$.

**Problem 10.** Find the number of integer solutions of $x_1 + x_2 + \cdots + x_5 = 30$, where $x_1 \ge 2,\ x_2 \ge 3,\ x_3 \ge 4,\ x_4 \ge 2,\ x_5 \ge 0$.

**Solution.** Let $y_1 = x_1 - 2;\ y_2 = x_2 - 3;\ y_3 = x_3 - 4;\ y_4 = x_4 - 2;\ y_5 = x_5$. Then $y_1, \ldots, y_5$ are non-negative integers. The equation becomes

$$(y_1 + 2) + (y_2 + 3) + (y_3 + 4) + (y_4 + 2) + y_5 = 30 \ \Rightarrow\ y_1 + y_2 + y_3 + y_4 + y_5 = 30 - 11 = 19.$$

Number of non-negative integer solutions $= C(5 + 19 - 1, 19) = C(23, 19) = 8855$.

**Problem 11.** How many integer solutions are there to $x_1 + x_2 + \cdots + x_5 = 20$, where $x_i \ge 2$?

**Solution.** Let $y_i = x_i - 2$ $(i = 1, \ldots, 5)$. Then $(y_1 + 2) + \cdots + (y_5 + 2) = 20 \Rightarrow y_1 + \cdots + y_5 = 10$. $\therefore$ Required number $= C(5 + 10 - 1, 10) = C(14, 10) = 1001$.

**Problem 12.** In how many ways can we distribute 12 identical pencils to 5 children so that every child gets at least 1 pencil?

**Solution.** First, distribute one pencil to each child. The remaining 7 pencils are to be distributed to 5 children $(n = 5,\ r = 7)$: $C(5 + 7 - 1, 7) = C(11, 7) = 330$ ways.

**Problem 13.** In how many ways can 10 identical pens be distributed among 5 children in the following cases: (i) there are no restrictions; (ii) each child gets at least one pen; (iii) the youngest child gets at least 2 pens?

**Solution.** (i) $C(5+10-1,10) = C(14,10) = 1001$.
(ii) First give one pen to each child; distribute the remaining 5 pens: $C(5+5-1,5) = C(9,5) = 126$.
(iii) First give 2 pens to the youngest child; distribute the remaining 8 pens: $C(5+8-1,8) = C(12,8) = 495$.

**Problem 14.** In how many ways can one distribute eight identical balls into 4 distinct containers so that (i) no container is empty? (ii) the 4th container gets an odd number of balls?

**Solution.** (i) First put one ball in each container. Distribute the remaining 4 balls among 4 containers: $C(4+4-1,4) = C(7,4) = 35$ ways.
(ii) The 4th container must get 1, 3, 5 or 7 balls.

- 1 ball in the 4th: remaining 7 balls into 3 containers: $C(3+7-1,7) = C(9,7)$.
- 3 balls: remaining 5 into 3 containers: $C(7,5)$.
- 5 balls: remaining 3 into 3 containers: $C(5,3)$.
- 7 balls: remaining 1 into 3 containers: $C(3,1)$.

$\therefore$ Total $= C(9,7) + C(7,5) + C(5,3) + C(3,1) = 36 + 21 + 10 + 3 = 70$.

**Problem 15.** Find the number of ways of distributing 7 identical pens and 7 identical pencils to 5 children so that each gets at least 1 pen and at least 1 pencil.

**Solution.** First give 1 pen and 1 pencil to each child. The remaining 2 pens can be distributed in $C(5+2-1,2) = C(6,2)$ ways, and similarly the remaining 2 pencils in $C(6,2)$ ways. $\therefore$ Number $= C(6,2) \times C(6,2) = 15 \times 15 = 225$.

**Problem 16.** Find the number of ways of giving 10 identical gift boxes to 6 persons A, B, C, D, E, F in such a way that the total number of boxes given to A and B together does not exceed 4.

**Solution.** Suppose $r$ boxes are given to A and B together; then $0 \le r \le 4$. Number of ways of giving $r$ boxes to A and B is

$$C(2+r-1,r) = C(r+1,r) = \frac{(r+1)!}{r!\,1!} = r+1.$$

Number of ways in which the remaining $(10-r)$ boxes can be given to C, D, E, F is $C\big(4+(10-r)-1,\ 10-r\big) = C(13-r,\ 10-r)$. Hence the total number of ways is

$$\sum_{r=0}^{4}(r+1)\times C(13-r,10-r) = 1(286)+2(220)+3(165)+4(120)+5(84) = 286+440+495+480+420 = 2121.$$

**Problem 17.** A total amount of Rs 1500 is to be distributed to 3 poor students A, B, C of a class. In how many ways can the distribution be made in multiples of Rs 100 if (i) every one of these must get at least Rs 300? (ii) A must get at least Rs 500, and B and C must get at least Rs 400 each?

**Solution.** Taking Rs 100 as a unit, there are 15 units for distribution among 3 students.
(i) First give 3 units to each of A, B, C; the remaining 6 units can be distributed in $C(3+6-1,6) = C(8,6) = 28$ ways.
(ii) First give 5 units to A and 4 units each to B and C; the remaining 2 units can be distributed in $C(3+2-1,2) = C(4,2) = 6$ ways.

**Problem 18.** A total of Rs 10,000 is to be distributed to 4 persons A, B, C, D in multiples of Rs 1000. In how many ways can the distribution be done (i) if there is no restriction? (ii) if every one of these persons should receive at least Rs 1000? (iii) if every one should receive at least Rs 1000 and A in particular should receive at least Rs 5000?

**Solution.** Taking Rs 1000 as one unit, there are 10 units for distribution among 4 persons.
(i) $C(4+10-1,10) = C(13,10) = 286$ ways.
(ii) Give 1 unit to each first; distribute the remaining 6 units: $C(4+6-1,6) = C(9,6) = 84$ ways.
(iii) Give 5 units to A and 1 unit each to B, C, D first; distribute the remaining 2 units: $C(4+2-1,2) = C(5,2) = 10$ ways.

**Problem 19.** A message is made up of 12 different symbols and is to be transmitted through a communication channel. In addition to the 12 symbols, the transmitter will also send a total of 45 blank spaces between the symbols, with at least 3 spaces between each pair of consecutive symbols. In how many ways can the transmitter send such a message?

**Solution.** The 12 symbols can be arranged in 12! ways. For each arrangement, there are 11 positions between the 12 symbols. Since there must be at least 3 spaces between consecutive symbols, we use up 33 of the 45 blank spaces. The remaining 12 blank spaces can be placed in the 11 positions in $C(11+12-1,12) = C(22,12)$ ways. $\therefore$ Number of ways

$$= C(22,12) \times 12! = \frac{22!}{10!\,12!} \times 12! = \frac{22!}{10!} \approx 3.0974 \times 10^{14}.$$

**Problem 20.** Six distinct symbols are transmitted through a communication channel. A total of 12 blanks are to be inserted between the symbols, with at least 2 blanks between every pair of symbols. In how many ways can the symbols and the blanks be arranged?

**Solution.** The 6 symbols can be arranged in 6! ways. There are 5 positions between them. At least 2 blanks in each uses up 10 of the 12 blanks. The remaining 2 blanks go into the 5 positions in $C(5+2-1,2) = C(6,2)$ ways. $\therefore$ Number of ways $= C(6,2) \times 6! = 15 \times 720 = 10{,}800$.

**Practice Questions — Combinations with Repetitions**

1. A sweet shop sells 4 kinds of sweets. In how many ways can a customer buy 10 sweets?
2. Find the number of positive integer solutions of $x_1 + x_2 + x_3 = 15$.
3. Find the number of distinct terms in the expansion of $(a+b+c)^{10}$.
4. Find the number of non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 \leq 10$.
5. Find the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 25$ with $x_1 \geq 2,\ x_2 \geq 1,\ x_3 \geq 0,\ x_4 \geq 5$.
6. In how many ways can 20 identical chocolates be given to 4 children if each child gets at least 3 and the eldest gets at most 8? (Hint: combine with Inclusion–Exclusion.)
7. How many solutions of $x_1 + x_2 + x_3 = 10$ in non-negative integers have $x_1 \leq 3$?

PART D
# Review

## 15 Unit 2 Question Bank

### Short-answer questions (2 marks)

1. State the Well-Ordering Principle. Give a set of numbers that is *not* well ordered.
2. State the Principle of Mathematical Induction and its alternate (strong) form.
3. Distinguish between the explicit and recursive definitions of a sequence, with an example.
4. State the Division Algorithm. Find $q$ and $r$ for $a = -23,\ b = 6$.
5. Define prime and composite numbers. Why is 1 not a prime?
6. State the Fundamental Theorem of Arithmetic and write 2520 in canonical form.
7. Define $\gcd(a, b)$ and state Bézout's identity.
8. Define $a \equiv b \pmod{m}$. Is $-7 \equiv 11 \pmod{9}$?
9. State the Principle of Inclusion–Exclusion for three sets.
10. State the Pigeon-hole Principle and its generalised form.
11. Distinguish between a permutation and a combination, with an example.
12. Write the number of non-negative integer solutions of $x_1 + x_2 + \cdots + x_n = r$.

### Long-answer questions (5–10 marks)

1. Prove by induction: $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \dfrac{n(n+1)(n+2)}{3}$.
2. Prove by induction that $3^{2n+2} - 8n - 9$ is divisible by 64 for all $n \geq 1$.
3. If $a_0 = 1,\ a_1 = 2$ and $a_n = 4a_{n-1} - 4a_{n-2}$, prove that $a_n = 2^n$ for all $n \geq 0$.
4. Prove the Division Algorithm using the Well-Ordering Principle.
5. Prove that there are infinitely many primes.
6. Using the Euclidean algorithm, find $\gcd(1819, 3587)$ and express it as $1819s + 3587t$.
7. Prove the Fundamental Theorem of Arithmetic.
8. Solve the linear congruence $7x \equiv 3 \pmod{15}$.
9. Find the number of integers between 1 and 1000 that are divisible by none of 2, 3, 5 and 7.
10. Find the number of derangements of $\{1, 2, 3, 4, 5, 6\}$.
11. Prove that if 101 integers are chosen from $\{1, 2, \ldots, 200\}$, then one of them divides another. (Hint: write each as $2^k \cdot m$, $m$ odd.)
12. How many arrangements of the letters of ENGINEERING have no two E's adjacent?
13. Find the coefficient of $x^2y^3z^4$ in $(2x - y + 3z)^9$.
14. Find the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 32$ with $x_i \geq 0$ and $x_1 \leq 10$.

# 16 Answers to Practice Questions

### Induction (Section 1.2)

Q1–Q9 are proofs. Hint for Q4: $A_{k+1} = 11A_k + 133 \cdot 12^{2k+1}$.

### Recursive Definitions

1. (i) $a_1 = 5,\ a_n = a_{n-1} + 3$ (ii) $a_1 = 1,\ a_n = a_{n-1} + 2n - 1$ (iii) $a_1 = 10,\ a_n = 2a_{n-1}$ (iv) $a_1 = 0,\ a_n = 2 - a_{n-1}$. 2. $a_n = 4n - 1$. 3. $a_n = 2 \cdot 3^n$. 4–7: proofs. 8. $1 \in S$; if $x \in S$ then $x + 2 \in S$.

### Division Algorithm

1. (i) $q = 34,\ r = 7$ (ii) $q = -8,\ r = 4$ (iii) $q = 0,\ r = 0$ (iv) $q = -1,\ r = 2$. 2. (i) 3 (ii) 4 (iii) $-8$. 3–6: proofs. 7. 1 o'clock ($109 \bmod 12 = 1$).

### Prime Numbers

1. $221 = 13 \cdot 17$, $437 = 19 \cdot 23$, 509 is prime, $1001 = 7 \cdot 11 \cdot 13$. 2. 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. 3. 25 primes. 4. Proof. 5. 3, 7, 11, 19, 23, 31, 43, 47. 6. 722, 723, 724, 725, 726.

### GCD and Euclidean Algorithm

1. (i) 139 (ii) 11 (iii) 20. 2. $8 = 4 \cdot 56 - 3 \cdot 72$. 3. $1 = (-9) \cdot 123 + 4 \cdot 277$. 4. 1,21,121. 5–7: proofs/program.

### Modular Arithmetic

1. (i) False (ii) False (iii) True (iv) True. 2. (i) 1 (ii) 1 (iii) 0. 3. 01. 4. (i) 7 (ii) 52 (iii) 34. 5. (i) $x \equiv 8 \pmod 9$ (ii) $x \equiv 67 \pmod{141}$. 6–7: proofs.

### Fundamental Theorem of Arithmetic

1. (i) $7 \cdot 11 \cdot 13$ (ii) $7^2 \cdot 11 \cdot 13$ (iii) $3 \cdot 11 \cdot 31$ (iv) $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. 2. $\tau(720) = 30, \sigma(720) = 2418$. 3. 58. 4. $\gcd = 2^3 \cdot 3^3 = 216$; $\operatorname{lcm} = 2^5 \cdot 3^5 \cdot 5^2 \cdot 7 = 13{,}60{,}800$. 5. Proof; $\sqrt{4} = 2$ — the parity argument fails because the exponents are already even. 6. 249. 7. Proof.

### Sum and Product Rules

1. 1024; 256. 2. $\sum_{k=6}^{8}(36^k - 26^k) = 2{,}68{,}44{,}83{,}06{,}33{,}360$. 3. 952. 4. 60. 5. 72; 90. 6. $n^m$; $n(n-1)\cdots(n-m+1) = P(n,m)$.

### Inclusion–Exclusion

1. 686. 2. 193; 57. 3. 52; 8. 4. 150; 8400. 5. $D_5 = 44,\ D_6 = 265$. 6. 161. 7. $\phi(360) = 96$. 8. $10! - 2 \cdot 9! + 8! = 29{,}43{,}360$.

### Pigeon-hole Principle

1. $\lceil 100/12 \rceil = 9$. 2. (i) 9 (ii) 42. 3. $36 \times 3 + 1 = 109$. 4. Remainders mod 6. 5. Holes $\{1,8\}, \{2,7\}, \{3,6\}, \{4,5\}$. 6. Divide into 4 triangles of side $\frac{1}{2}$. 7. Holes $\{1,2\}, \{3,4\}, \ldots, \{2n-1, 2n\}$. 8. 60 cables. 9. Erdős–Szekeres theorem.

### Permutations

1. $n = 9$. 2. $\dfrac{13!}{3!\,4!\,2!\,2!} = 1{,}08{,}10{,}800$. 3. $6! \times 3! = 4320$. 4. 720; $6! - 2 \cdot 5! = 480$. 5. $3 \times 4! = 72$. 6. $6! = 720$.

### Combinations

1. (i) 560 (ii) 686. 2. (i) 66 (ii) 220. 3. $\dfrac{n(n-3)}{2}$; $n = 5$. 4. 56; $1 + 8 + 28 = 37$. 5. 276. 6. Proof.

### Binomial and Multinomial Theorems (Homework)

1. 6048. 2. 35. 3. $-30{,}24{,}000$. 4. $1{,}61{,}280$. 5. 12,600. 6. Proof.

### Combinations with Repetitions

1. $C(13, 10) = 286$. 2. $C(14, 2) = 91$. 3. $C(12, 2) = 66$. 4. $C(14, 10) = 1001$. 5. $C(20, 3) = 1140$. 6. $C(11, 3) - C(5, 3) = 165 - 10 = 155$. 7. $C(12, 2) - C(8, 2) = 66 - 28 = 38$.

**End of Unit 2**